MICROSOFT

POWER PLATFORM

with Copilot

COURSE SLIDES · WSQ

Create Intelligent Power Apps and Power Automate Workflows with Copilot

WSQ Course Code: TGS-2023036648

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera (or Singpass app) and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

CERTIFICATION
Microsoft Power Platform (PL-100 / PL-400), Azure and applied AI — 20+ years of training experience.

DELIVERS
WSQ courses on Power Platform, business automation, AI and data analytics.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

1

Who you are

Your name, organisation and role.

2

Your starting point

Your experience with Power Apps, Power Automate, Excel or any low-code tool.

3

The process you'd automate

One business process in your organisation you would most like to automate.

4

What you want to leave with

What you want to walk out of this 4-day course able to build.

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HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Respect each other's views; agree to disagree.
4. One conversation at a time.
5. Exit quietly for a call or a break, and return on time.
6. 75% attendance is required for funding.

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LMS / TMS · lms-tms.tertiaryinfotech.com

Download Your Course Material

1

1 · Go to the LMS portal

Open https://lms-tms.tertiaryinfotech.com in your browser.

2

2 · Log in

Sign in with the account e-mail you used to register for this course.

3

3 · Open this course

Select 'Create Intelligent Power Apps and Power Automate Workflows with Copilot'.

4

4 · Download

Download the Trainer Slides, Learner Guide and Lesson Plan (PDF).

5

5 · Get the labs

Open the labs folder — 26 labs, each with its own step-by-step guide and mock data.

6

6 · Keep them open

You may use these materials during the open-book assessment.

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COURSE ENVIRONMENT

Your Training Environment

1

Microsoft 365 account

The trainer assigns you a training account — use the same login for every lab.

2

One environment

Stay in the environment the trainer names (e.g. Dev One). Flows, apps and agents must share it.

3

Power Apps

make.powerapps.com — canvas apps, model-driven apps, Dataverse tables.

4

Power Automate

make.powerautomate.com — cloud flows, approvals, run history.

5

Copilot Studio

copilotstudio.microsoft.com — agents, agent flows, knowledge and tools.

6

Do not change passwords

These are shared training accounts. Leave the credentials as issued.

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Skills Framework Alignment

1

TSC Title

Applications Integration

2

TSC Code

ICT-DIT-3003-1.1

3

A1 / A2

Identify integration opportunities across devices, databases and applications; perform feasibility scans.

4

A3 / A4

Utilise middleware to integrate data and functions across programs; support API-level integration.

5

A5 / A6

Perform tests and checks on connections; verify modules function across integrated platforms.

6

A7 / A8

Highlight technical, compatibility and performance issues; implement modifications to middleware.

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · Identify & scan

Identify opportunities for creating connections in Microsoft Power Platform and perform feasibility scans.

2

LO2 · Integrate with Power Apps

Utilise Power Apps to integrate data and functions across programs and API-level integration.

3

LO3 · Test with Power Automate

Execute tests in Power Automate to verify connections between disparate programs across platforms.

4

LO4 · Resolve & modify

Resolve technical issues in Power Platform and implement modifications to Power Platform middleware.

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19 TOPICS · 26 HANDS-ON LABS

Course Outline — Four Learning Units

1

LU1 — Get Started with Copilot for Power Platform

Topics 1–4 · Copilot in Power Apps, Power Automate and Power Pages · Labs 1–4.

2

LU2 — Create and Manage Canvas Apps

Topics 5–10 · data, formulas, UI, external data, forms, publishing, PCF · Labs 5–12.

3

LU3 — Automated Processes with Power Automate & Copilot Studio Agents

Topics 11–14 · flows, approvals, filters, agents, tools, structured output · Labs 13–20.

4

LU4 — Model-Driven Apps with Power Apps and Dataverse

Topics 15–19 · tables, forms, views, charts, solutions, troubleshooting · Labs 21–26.

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SCHEDULE · 9:30am – 6:30pm

Lesson Plan — Days 1 & 2

Day 1

- Day 1 — Copilot for Power Platform & First Canvas Apps

– Digital Attendance (AM) · Introductions · Learning Outcomes

– LU1 Topics 1–4: Copilot across Power Platform

– Labs 1–4: environment scan, Copilot app, Copilot flow, Power Pages

– Lunch Break · Digital Attendance (PM)

– LU2 Topics 5–6: canvas apps and the Dataverse data model

– Labs 5–7: Excel app, Dataverse tables, canvas app over Dataverse

Day 2

- Day 2 — Canvas Apps: UI, External Data, Forms & Publishing

– Digital Attendance (AM) · Day 1 recap

– LU2 Topics 7–8: building the UI and working with external data

– Labs 8–9: theming and branding, SharePoint + collections + Patch

– Lunch Break · Digital Attendance (PM)

– LU2 Topics 9–10: writing data with forms, publish and maintain

– Labs 10–12: forms, PCF code component, publish and share

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SCHEDULE · 9:30am – 6:30pm

Lesson Plan — Days 3 & 4

Day 3

- Day 3 — Power Automate Flows & Copilot Studio Agents

– Digital Attendance (AM) · Day 2 recap

– LU3 Topics 11–12: flow anatomy, triggers, actions, approvals

– Labs 13–15: three ways to build a flow, approvals, button & scheduled

– Lunch Break · Digital Attendance (PM)

– LU3 Topics 13–14: trigger filters, Copilot Studio agents

– Labs 16–20: filters, agents, tools, structured output, multi-agent

Day 4

- Day 4 — Model-Driven Apps, Dataverse, Solutions & Assessment

– Digital Attendance (AM) · Day 3 recap

– LU4 Topics 15–17: Dataverse tables, model-driven apps, forms & views

– Labs 21–24: tables, model-driven app, forms/views, charts

– Lunch Break · Digital Attendance (PM)

– LU4 Topics 18–19: solutions, ALM and troubleshooting · Labs 25–26

– Briefing · Digital Attendance (Assessment) · WA + PP · TRAQOM

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1

LU1

Get Started with Copilot for Microsoft Power Platform

Topics 1–4 · Labs 1–4

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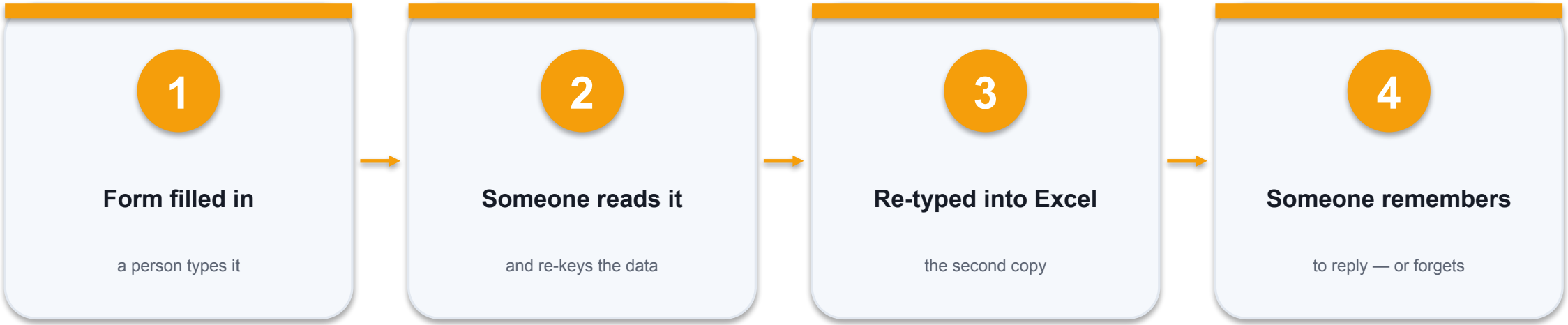
CORE CONCEPTS

The Power Platform, and What Copilot Changes

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CORE CONCEPT

What Business Process Automation Removes



THE HAND-OFF IS THE COST

Automation is not software doing the same work faster. It removes the hand-off where a person re-types what another system already knows.

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CORE CONCEPT · THE PLATFORM

The Power Platform — Four Products, One Environment

1

Power Apps

Canvas and model-driven applications over your data. Labs 2, 5–12, 21–24.

2

Power Automate

Cloud flows: triggers and actions that run without a person. Labs 3, 13–16.

3

Copilot Studio

Conversational agents with instructions, knowledge and tools. Labs 17–20.

4

Power Pages

External-facing websites backed by Dataverse. Lab 4.

5

Dataverse

The managed data store behind the environment. Labs 6, 21, 25.

6

Connectors

Excel, SharePoint, Outlook, Teams, SQL, HTTP — 1,000+ integration points.

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CORE CONCEPT · ENVIRONMENTS

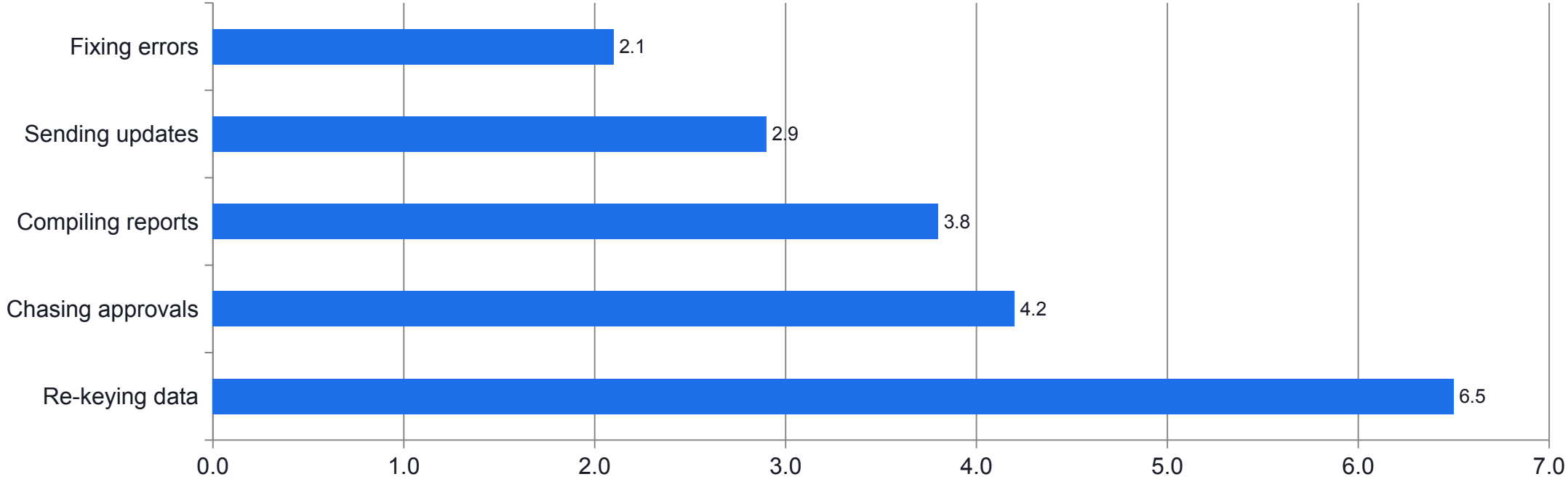
The environment is the container.

Apps, flows, agents and data all live inside ONE environment. If Power Automate and Copilot Studio point at different environments, your agent cannot see your flow. This is the single most common "where did it go?" in this course.

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CORE CONCEPT · THE CASE FOR AUTOMATION

Where the Manual Hours Actually Go



WHAT THE DATA SHOWS

Re-keying and chasing approvals — the two hand-offs a flow removes outright — account for more than half the manual hours. These map directly to Labs 13–16.

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A Copilot Is an AI Assistant You Reach Through Chat

1

Generic copilot

Microsoft Copilot — not tied to any one product.

2

Context-aware copilot

Tailored to the product you are in right now.

3

Maker-facing

Helps you BUILD the app, flow or site — Labs 2, 3, 4.

4

User-facing

Helps the END USER ask questions of the app's data.

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1

TOPIC 01

Introduction to Copilot in Microsoft Power Platform

What a copilot is · the copilots in Power Platform · anatomy · prompts

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TOPIC 1 · CONCEPT

The Copilots Across Microsoft Power Platform

Copilot	What you give it	What it produces	Used in
Copilot for Power Apps	A description of the app you need	A data structure plus canvas app pages you can iterate	Lab 2
Copilot for Power Automate	A description of the automation	A cloud or desktop flow — explains, adds, replaces actions	Lab 3
Copilot for Power Pages	A description of the website	Theming, standard pages, AI images and text	Lab 4
Copilot in Power BI	A question about the data	Visuals, DAX measures and narrative summaries	Reference
User-facing Copilot	A question about the app's records	A natural-language answer over the app's data	Lab 2

WHEN IT MATTERS Every one of these is an accelerator, not an author. The maker still owns correctness — Copilot's proposed data structure is a starting point you review and correct.

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The Anatomy of a Copilot — Four Things Every One Shares

1

Compose a prompt

Ask the copilot to take an action, in natural language.

2

Undo modifications

Roll back a change that did not work out.

3

Ask for help

Including asking the copilot what it can actually do.

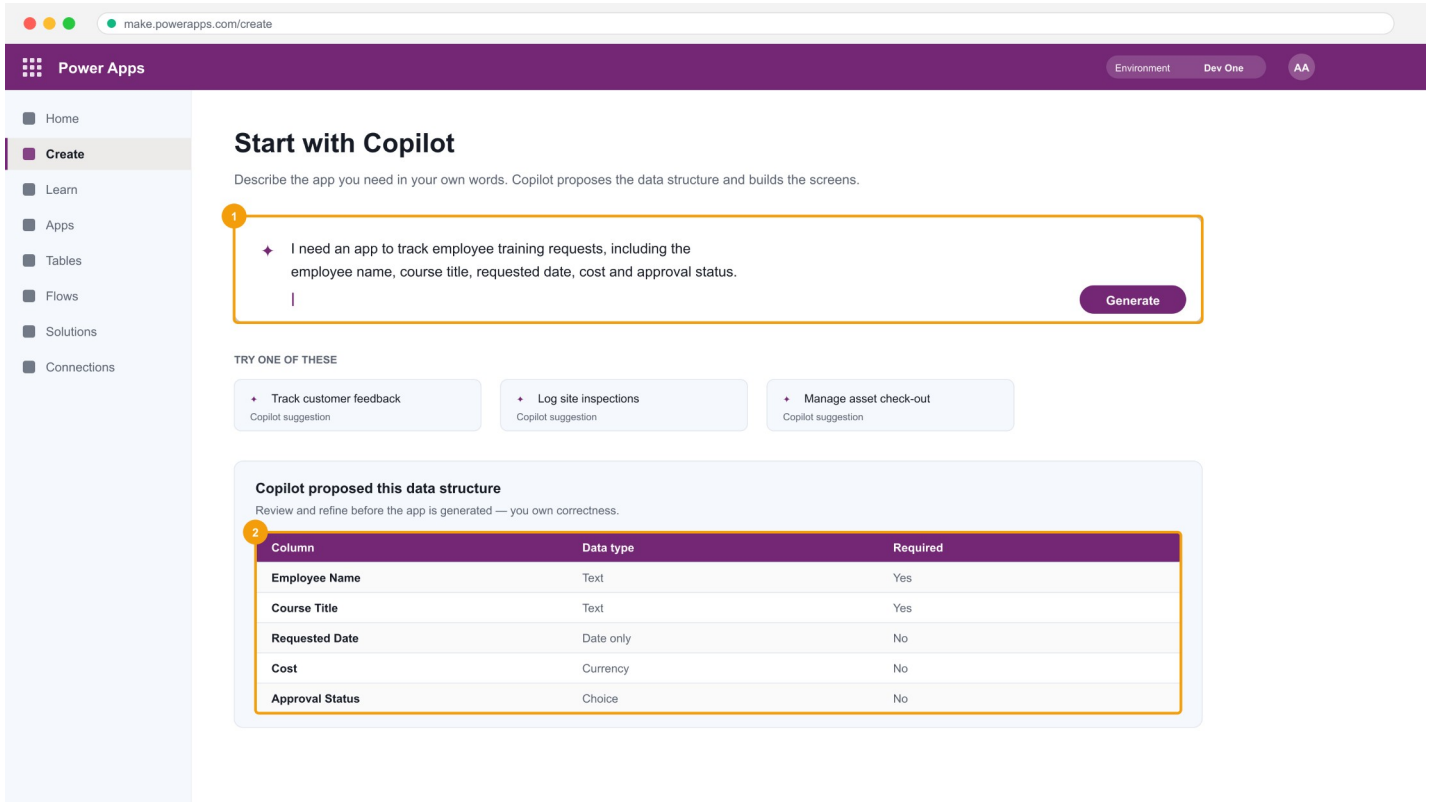
4

Show or hide

Make the copilot pane visible or hidden beside your work.

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Copilot in Power Apps — Describe It to Design It



1. The prompt box. A natural-language description of the outcome you want.

2. The proposed data structure. Review the column names and data types, and refine by prompting again — before the app is generated.

Applied in Lab 2 — you will describe an app, refine the data structure by prompt, then generate and inspect it.

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What Makes a Prompt Work

1

Be specific

"Track employee training requests" beats "an HR app".

2

Name the fields

List the columns you know you need, and their meaning.

3

State the outcome

Describe what the app must let someone DO.

4

Iterate, don't restart

Refine with a follow-up prompt rather than starting over.

5

Review everything

Copilot proposes; you verify names, types and relationships.

6

Know the goal first

If you cannot state the outcome, the copilot cannot either.

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Feasibility Scan — Can This Integration Actually Be Built?



YES — build it on the connector

Use the standard connector. Confirm the action you need exists, then check throttling limits and the data volume.

NO — escalate the design

Consider a premium connector (licence cost), a custom connector over the vendor's API, or HTTP with authentication.

This decision is the TSC A2 deliverable. In Lab 1 you record the result for ten connectors in connector-scan.csv — that file is your evidence of a feasibility scan.

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TOPIC 1 · THE HONEST FRAMING

Copilot accelerates. It does not absolve.

Prototyping, inspiration and task completion are real gains. But the maker still owns the data model, the correctness of the formulas and the security of the result — and that is exactly what the assessment tests.

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2

TOPIC 02

Use Copilot in Power Apps

Maker-facing Copilot · user-facing Copilot · what it generates · what you must review

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TOPIC 2 · CONCEPT

Maker-Facing and User-Facing Copilot Are Different Things

	Maker-facing Copilot	User-facing Copilot
Who uses it	You, while BUILDING the app	The end user, while USING the app
What it does	Proposes a data structure, generates screens, edits controls	Answers questions about the data in the app
Where it appears	The maker portal and Power Apps Studio	A Copilot control on a screen
What it changes	The app itself	Nothing — it reads and answers
Built in	Lab 2	Lab 2 (optional extension)

WHEN IT MATTERS Adding the user-facing Copilot control lets a user ask 'how many requests are still pending?' in natural language, over the same data the gallery shows.

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What Copilot Generates — and What You Must Check

1

A data structure

Column names and DATA TYPES. Check Currency is Currency, dates are Date only, and choices are Choice.

2

Screens and controls

A browse, detail and edit screen. Check the gallery's Items and the form's DataSource.

3

Power Fx formulas

It writes real formulas. Check delegation, and that filters use the right column.

4

Nothing about security

Copilot does NOT set security roles or sharing. That remains entirely yours.

5

Nothing about performance

It will happily generate a non-delegable filter. You own the fix.

6

A starting point

Every generated artefact is a draft you review, correct and extend.

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TOPIC 03

Use Copilot in Power Automate

Describing a flow · explaining actions · adding and replacing actions

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TOPIC 3 · CONCEPT

What Copilot for Power Automate Can Actually Do

1

Build from a description

'When a new item is added to my list, email me the details' becomes a real flow with a trigger and actions.

2

Explain an action

Ask what a step does and why it is there — useful on a flow someone else built.

3

Add an action

'Also post it to Teams' inserts the action in the right place, already wired to the earlier outputs.

4

Replace an action

Swap one connector action for another without rebuilding the surrounding steps.

5

Answer questions

Ask about connectors, expressions or why a run failed.

6

NOT design the process

It automates the process you describe. Deciding what SHOULD be automated is still your judgement.

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TOPIC 3 · THE JUDGEMENT THAT STAYS YOURS

Copilot builds the flow. You still own the trigger.

The trigger is a statement about who or what may start a business process — and that is a design decision, not a generation task. Read what Copilot chose, and change it if the wrong thing is allowed to start the process.

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4

TOPIC 04

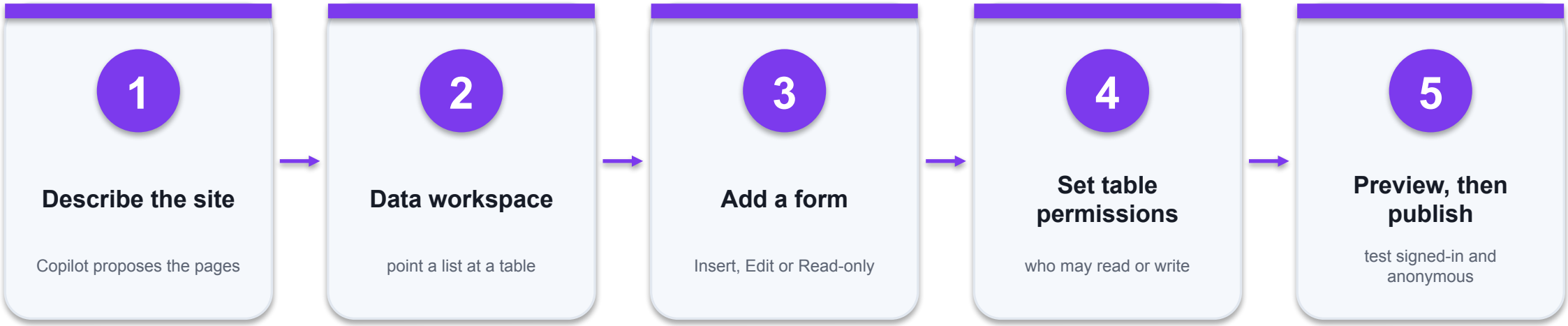
Use Copilot in Power Pages

Generating a site · theming and pages · Dataverse-backed lists and forms · permissions

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TOPIC 4 · CONCEPT

How a Power Pages Site Reaches Real Data



PERMISSIONS ARE THE WHOLE POINT

A Power Pages site is EXTERNAL-facing. Table permissions and web roles decide what an anonymous visitor can see — get this wrong and you have published your Dataverse to the internet.

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What Copilot Generates for a Power Pages Site

1. **Theming**
   A colour scheme and typography you can then edit.

2. **Standard pages**
   Home, about and contact pages as a starting structure.

3. **AI-generated images**
   Stock imagery matched to your description.

4. **Text content**
   Headings and descriptive copy — review it for accuracy before publishing.

5. **NOT the data binding**
   You add Dataverse lists and forms yourself, in the Data workspace.

6. **NOT the permissions**
   Table permissions and web roles are always your decision.

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WHAT YOU'LL DO

Hands-On Labs — Get Started with Copilot for Microsoft Power Platform

Labs 1–2

- Validate the Lab Environment and Explore Copilot
- Use Copilot to Build a Canvas App from a Description

Labs 3–4

- Build a Cloud Flow with Copilot for Power Automate
- Use Copilot to Build a Power Pages Site

—

- —

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LU1 · HANDS-ON

Validate the Lab Environment and Explore Copilot

LAB 1

A1/A2 — Identify connection opportunities in Power Platform and scan the environment for feasibility.

Sign in to the Power Apps maker portal, confirm you are in the correct environment, verify that Dataverse and the Copilot features are available, and take an inventory of the connectors your tenant can actually reach.

YOU'LL BUILD

A validated maker environment with a documented connector feasibility scan (mock-data/connector-scan.csv).

TOOLCHAIN

Power Apps maker portal, Power Platform admin centre, Dataverse, Copilot

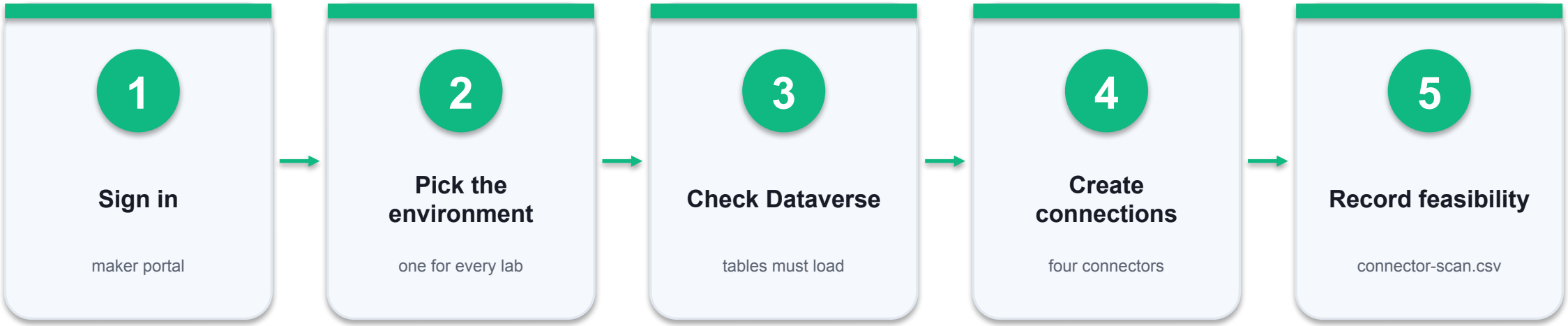
DONE WHEN

Tables load in the chosen environment, all four connections show Connected, and connector-scan.csv records the feasibility result for each.

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LAB 1 · PROCESS MAP

How Lab 1 Runs



THE POINT

A1/A2 — Identify connection opportunities in Power Platform and scan the environment for feasibility.

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LAB 1 · VERIFY

Validate the Lab Environment and Explore Copilot

✅ Expected result

Tables load in the chosen environment, all four connections show Connected, and connector-scan.csv records the feasibility result for each.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-01/README.md.

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LU1 · HANDS-ON

Use Copilot to Build a Canvas App from a Description

LAB 2

A1/A3 — Use Copilot to turn a natural-language need into a working app over a generated data structure.

Describe an app in plain English and let Copilot propose a data structure and screens. Iterate on the proposed table, then let Copilot generate the app and inspect what it produced.

YOU'LL BUILD

A Copilot-generated canvas app over a Dataverse table, with the generated data structure reviewed and corrected.

TOOLCHAIN

Power Apps, Copilot for Power Apps, Dataverse

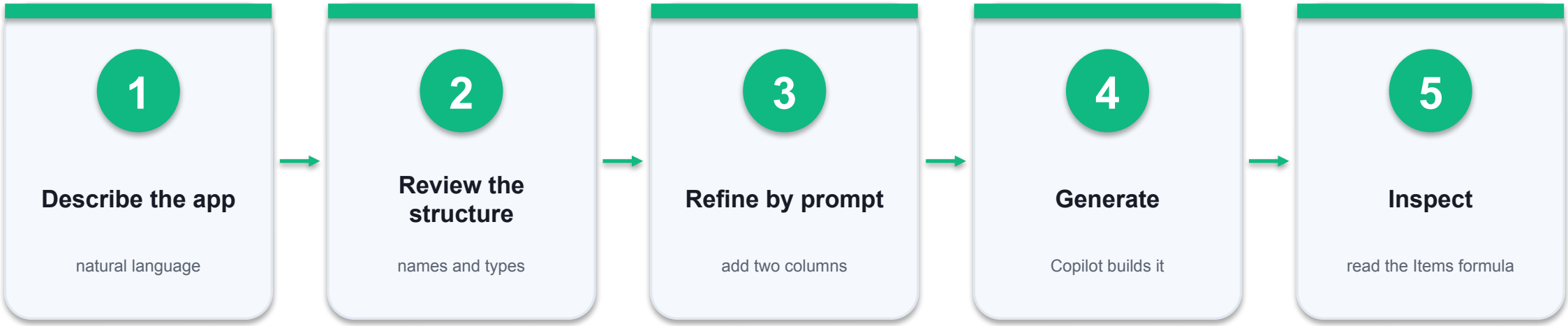
DONE WHEN

The app opens in Preview, a record added in the form appears in the gallery, and the Dataverse table contains the two columns you added by prompt.

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LAB 2 · PROCESS MAP

How Lab 2 Runs



THE POINT

A1/A3 — Use Copilot to turn a natural-language need into a working app over a generated data structure.

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LAB 2 · VERIFY

Use Copilot to Build a Canvas App from a Description

✅ Expected result

The app opens in Preview, a record added in the form appears in the gallery, and the Dataverse table contains the two columns you added by prompt.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-02/README.md.

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LU1 · HANDS-ON

Build a Cloud Flow with Copilot for Power Automate

LAB 3

A3/A5 — Use Copilot to assemble a flow, then read and test what it built.

Use Copilot in Power Automate to build a flow from a description, then examine the trigger and actions it chose, correct them, and test the run.

YOU'LL BUILD

A working Copilot-built cloud flow that emails a summary when a new item is created.

TOOLCHAIN

Power Automate, Copilot for Power Automate, Outlook, SharePoint

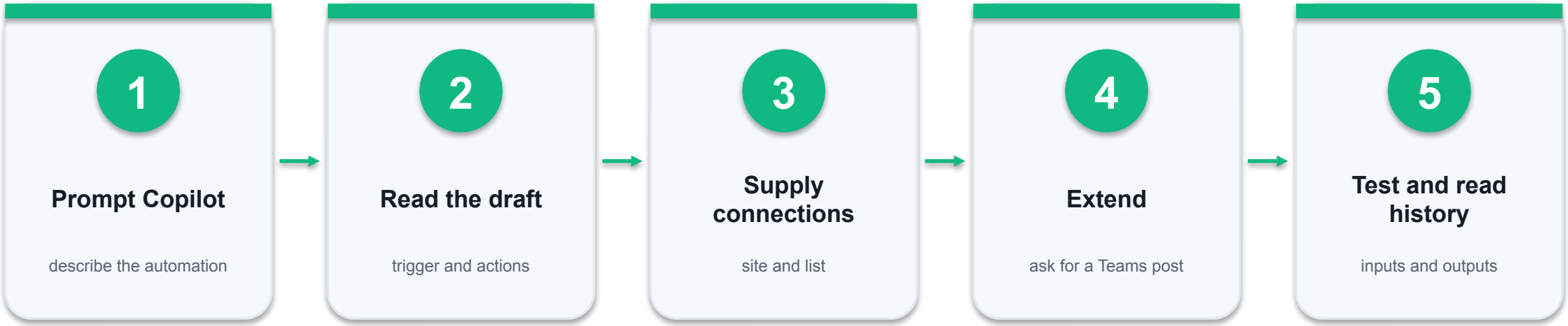
DONE WHEN

The run history shows a Succeeded run, the email arrives with the item's field values, and you can point at the action output that supplied each value.

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LAB 3 · PROCESS MAP

How Lab 3 Runs



THE POINT

A3/A5 — Use Copilot to assemble a flow, then read and test what it built.

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LAB 3 · VERIFY

Build a Cloud Flow with Copilot for Power Automate

✅ Expected result

The run history shows a Succeeded run, the email arrives with the item's field values, and you can point at the action output that supplied each value.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-03/README.md.

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LU1 · HANDS-ON

Use Copilot to Build a Power Pages Site

LAB 4

A1/A4 — Generate an external-facing site and connect it to Dataverse data through its API layer.

Describe a public-facing site to Copilot in Power Pages, review the theming, pages and AI-generated content it produces, then add a Dataverse-backed form so the site reads and writes real data.

YOU'LL BUILD

A published Power Pages site with a Dataverse-backed page and a working form.

TOOLCHAIN

Power Pages, Copilot for Power Pages, Dataverse

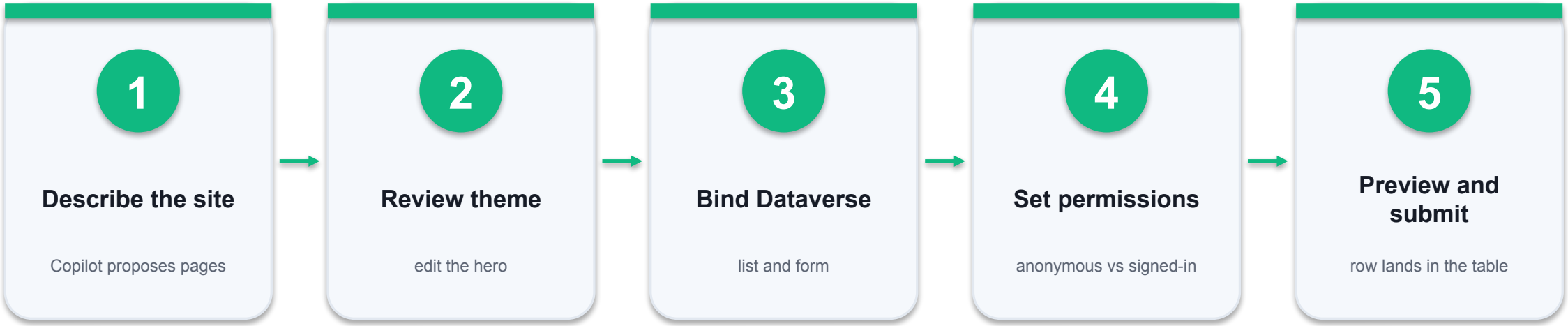
DONE WHEN

The previewed site lists courses from Dataverse and a submitted request creates a new row in the Dataverse table.

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LAB 4 · PROCESS MAP

How Lab 4 Runs



THE POINT

A1/A4 — Generate an external-facing site and connect it to Dataverse data through its API layer.

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LAB 4 · VERIFY

Use Copilot to Build a Power Pages Site

✅ Expected result

The previewed site lists courses from Dataverse and a submitted request creates a new row in the Dataverse table.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-04/README.md.

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LEARNING UNIT RECAP · YOU CAN NOW

Recap — Get Started with Copilot for Microsoft Power Platform

1

Lab 1 — Validate the Lab Environment and Explore Copilot

A1/A2 — Identify connection opportunities in Power Platform and scan the environment for feasibility.

2

Lab 2 — Use Copilot to Build a Canvas App from a Description

A1/A3 — Use Copilot to turn a natural-language need into a working app over a generated data structure.

3

Lab 3 — Build a Cloud Flow with Copilot for Power Automate

A3/A5 — Use Copilot to assemble a flow, then read and test what it built.

4

Lab 4 — Use Copilot to Build a Power Pages Site

A1/A4 — Generate an external-facing site and connect it to Dataverse data through its API layer.

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2

LU2

Create and Manage Canvas Apps with Power Apps

Topics 5–10 · Labs 5–12

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5

TOPIC 05

Get Started with Power Apps Canvas Apps

What a canvas app is · Power Fx · controls · galleries and forms

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TOPIC 5 · CONCEPT

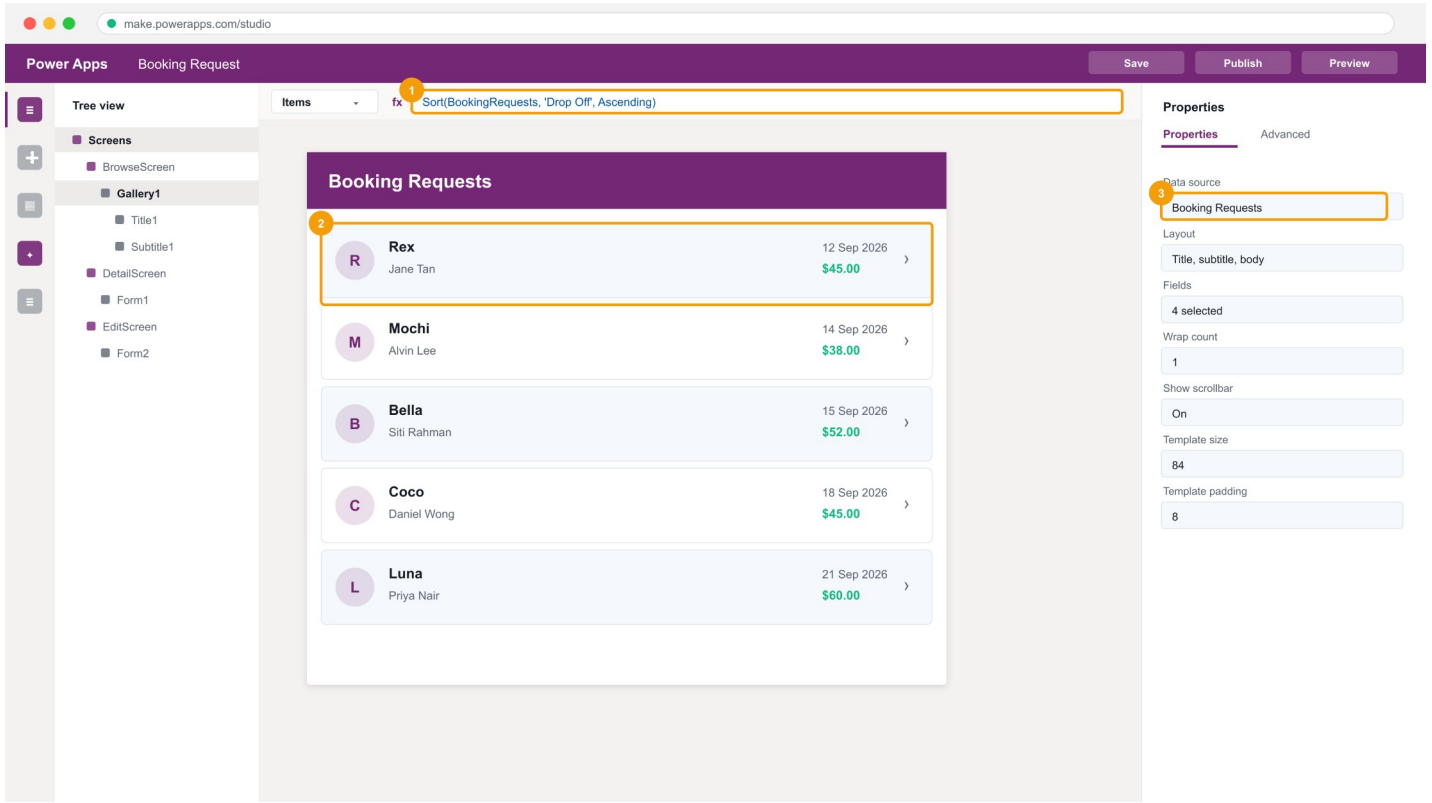
Canvas App or Model-Driven App — Which One?

	Canvas app	Model-driven app
You design	Pixel-precise screens you draw	Components generated from the data model
Starts from	A blank screen or a data source	The Dataverse tables themselves
Data source	Any of 1,000+ connectors	Dataverse only
Layout control	Total — you place every control	Configured, not drawn
Best for	Task-focused, mobile, branded UX	Data-heavy, process-driven business apps
In this course	Labs 5–12	Labs 21–24

WHEN IT MATTERS Neither is 'better'. Choose canvas when the experience matters most; choose model-driven when the data model and process matter most. LU2 builds one of each.

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Power Apps Studio — Where You Will Spend LU2



1. The formula bar. Select a property on the left, then write the Power Fx expression that produces its value.
2. The gallery — one control that repeats a template for every record in its Items collection.
3. The properties pane. Everything here can also be set as a formula in the bar above.

Three panes: the tree view of every control, the canvas you are designing, and the properties of whatever is selected.

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Power Fx — A Property Is an Expression, Not a Script

Power Fx is declarative and recalculates automatically, exactly like an Excel cell. There is no 'run' — you never write a loop to fill a gallery.

```
// Gallery1.Items – the rows to show
Sort(
    Filter('Booking Requests',
           Status.Value = "Requested"),
    'Drop Off', Ascending
)

// Label1.Text – recalculates by itself
"Showing " & CountRows(Gallery1.AllItems)
         & " pending requests"

// Button.OnSelect – actions, in order
Patch('Booking Requests',
      Gallery1.Selected,
      {Status: {Value: "Confirmed"}});
Notify("Booking confirmed", Success)
```

Declarative

You state what a property IS. When the data changes, every dependent property recalculates on its own.

Filter then Sort

Filter narrows the rows, Sort orders them. Both should be delegable so the server does the work.

& concatenates

One expression can mix literals, functions and control properties.

; sequences behaviour

Only behaviour properties (OnSelect, OnVisible) take a ;-separated sequence.

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The Controls You Will Actually Use

1 **Label**
Displays text. Its Text property is a formula.

2 **Gallery**
Repeats a template for every record in Items.

3 **Form (Display / Edit)**
Shows or edits ONE record — the Item property.

4 **Text input**
Collects typed input; read it back as .Text.

5 **Button / Icon**
Runs a behaviour formula in OnSelect.

6 **Dropdown / Combo box**
Choose from a list or a related table.

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6

TOPIC 06

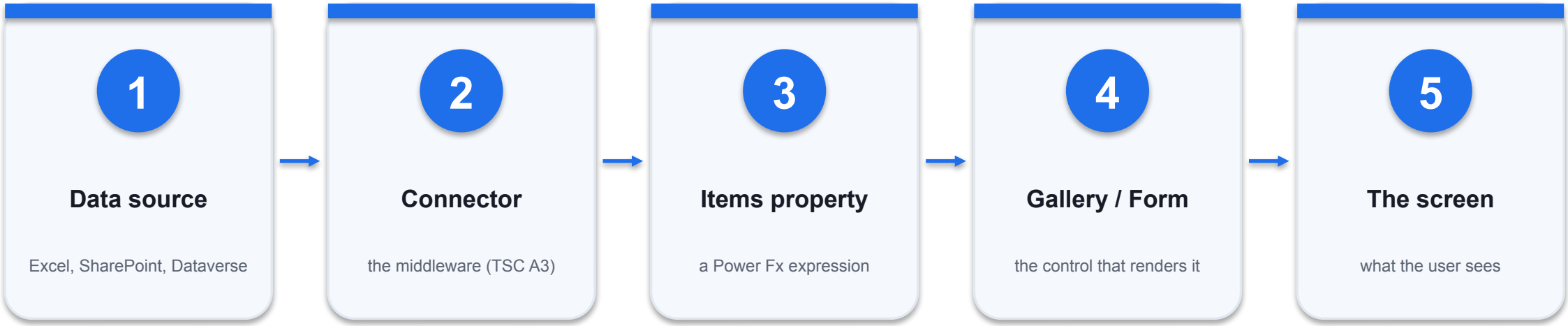
Customize a Canvas App in Power Apps

Formulas · screen navigation · the data model · variables and collections

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TOPIC 6 · CONCEPT

How Data Reaches the Screen



THE CHAIN IS THE INTEGRATION

Every canvas app is this chain. When data does not appear, you debug the chain one link at a time — which is exactly the Lab 26 method.

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TOPIC 6 · STATE

Where to Keep a Value — Three Kinds of State

	Global variable	Context variable	Collection
Set with	Set(gblUser, …)	UpdateContext({locX: …})	ClearCollect(colX, …)
Scope	The whole app	One screen only	The whole app
Holds	A single value or record	A single value or record	A TABLE of records
Survives navigation	Yes	No	Yes
Typical use	Signed-in user, app settings	Show/hide a panel on this screen	Offline cache, staged edits
Lab	Lab 8	Lab 10	Lab 9

WHEN IT MATTERS Reach for a variable only when a formula cannot express it directly. Power Fx recalculates automatically — an unnecessary variable is state you now have to keep correct yourself.

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Navigation Between Screens

1

Navigate(Screen2)

Go to another screen.

2

Navigate(Screen2, ScreenTransition.Cover)

Go, with a transition.

3

Navigate(Screen2, None, {locId: ThisItem.ID})

Go, passing context.

4

Back()

Return to the previous screen — remembers where you came from.

5

Gallery1.Selected

The record the user picked — how a detail screen knows what to show.

6

NewForm(Form1); Navigate(EditScreen)

Open a blank form to create a record.

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7

TOPIC 07

How to Build the User Interface in a Canvas App

Themes · branding · fonts and icons · personalisation · responsive layout

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TOPIC 7 · CONCEPT

Building a UI People Can Actually Use

1

Theme first, then override

Settings › Theme restyles every control at once. Override individual controls afterwards.

2

Font

Available on any control that displays text — 14 built-in fonts, or type any font name.

3

Icons over labels

An icon needs no text and fits where a label will not — the floppy disk for Save, the arrow for Back.

4

Colour with meaning

Use colour to signal state (Confirmed / Declined), not decoration.

5

Personalise

User().FullName and Office365Users.UserPhotoV2() greet the signed-in person.

6

Design for the target

Tablet and phone layouts differ. Set X/Y and Width as formulas so the layout holds.

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8

TOPIC 08

Work with External Data in a Power Apps Canvas App

Connectors · SharePoint · collections · Patch · delegation

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Connectors Are the Middleware (TSC A3)

1

Excel Online (Business)

Binds to a NAMED TABLE inside a workbook in OneDrive. Lab 5.

2

SharePoint

Lists behave like tables; choice columns need {Value: "…"}. Lab 9.

3

Dataverse

The native store — relationships, choices, security roles. Labs 6, 7, 10.

4

Office 365 Users

Profile, photo and manager of the signed-in user. Lab 8.

5

SQL Server (premium)

Direct table access — check the licence before you design on it.

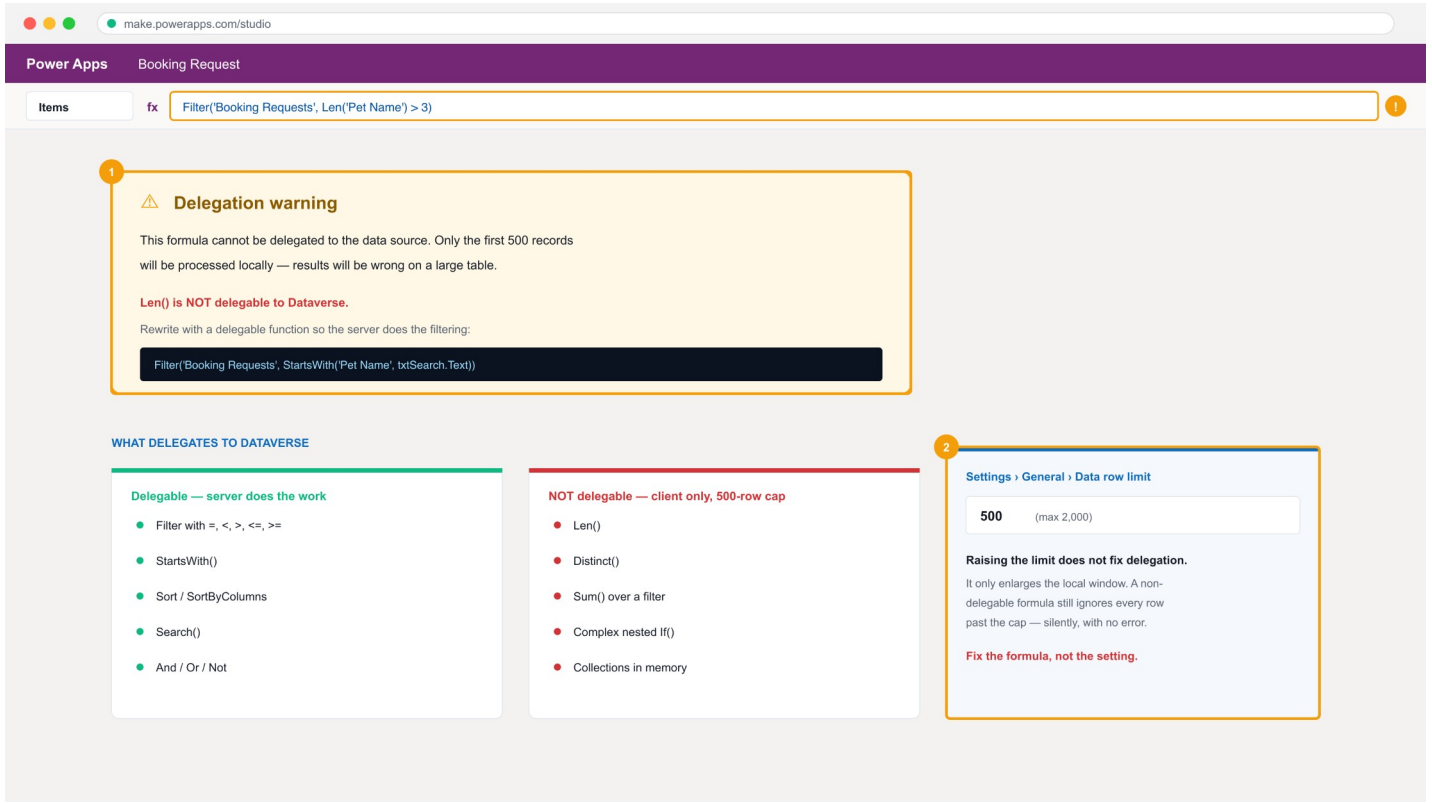
6

Custom connector / HTTP

When no standard connector exists — the TSC A4 API-level path.

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Delegation — The Silent Failure You Must Recognise



1. The blue warning. The formula cannot be delegated, so only the first 500 rows are processed — locally.

2. Raising the data row limit does NOT fix it. It only enlarges the local window. Fix the formula instead.

A non-delegable formula does not error. It silently returns wrong answers as soon as the table outgrows the cap — the classic TSC A7 performance issue.

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Writing Data — Patch, SubmitForm and Collections

Three ways to write. Choose by how much control you need over the record and the moment of the write.

```
// 1 — Patch: write specific fields
Patch('Kennel Bookings',
      ThisItem,
      {Status: {Value: "Confirmed"}})

// 2 — Patch: create a NEW record
Patch('Booking Requests',
      Defaults('Booking Requests'),
      {'Pet Name': txtPet.Text,
       'Owner Name': txtOwner.Text})

// 3 — SubmitForm: commit a whole Form
SubmitForm(Form2)

// Load rows into memory (client-side)
ClearCollect(colBookings,
             'Kennel Bookings')
```

Patch updates

Give it the data source, the record and only the fields to change.

Defaults() creates

Patching over Defaults() inserts a new row instead of updating one.

SubmitForm commits

The Form validates required fields first, then writes — use OnSuccess to navigate.

ClearCollect caches

A collection is a client-side table. It does not write back until you Patch it.

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9

TOPIC 09

Write Data in a Power Apps Canvas App

Form controls · SubmitForm · validation · errors · LastSubmit

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The Form Control — Modes and What Each One Does

Mode	Set by	What the user sees	Commit with
FormMode.View	ViewForm(Form1)	Read-only display of the record	— (nothing to commit)
FormMode.Edit	EditForm(Form1)	Editable fields for an EXISTING record	SubmitForm(Form1)
FormMode.New	NewForm(Form1)	Blank fields for a NEW record	SubmitForm(Form1)

WHEN IT MATTERS A Form's Item property says WHICH record; its Mode says what the user may do with it. Set Mode with the function, then let OnSuccess navigate away and OnFailure show Form1.Error.

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Verifying a Write Actually Happened (TSC A5)

1

Form1.Error

The error message text from the last failed submit — put it in a Label.

2

Form1.LastSubmit

The whole record as written, including the server-generated ID.

3

OnSuccess

Fires only when the write succeeded — the right place to Back() or Notify().

4

OnFailure

Fires when it did not — show the error rather than failing silently.

5

Check the source

Open the Dataverse table or SharePoint list and confirm the row changed.

6

Required columns

A Business-required column blocks the submit until it has a value.

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10

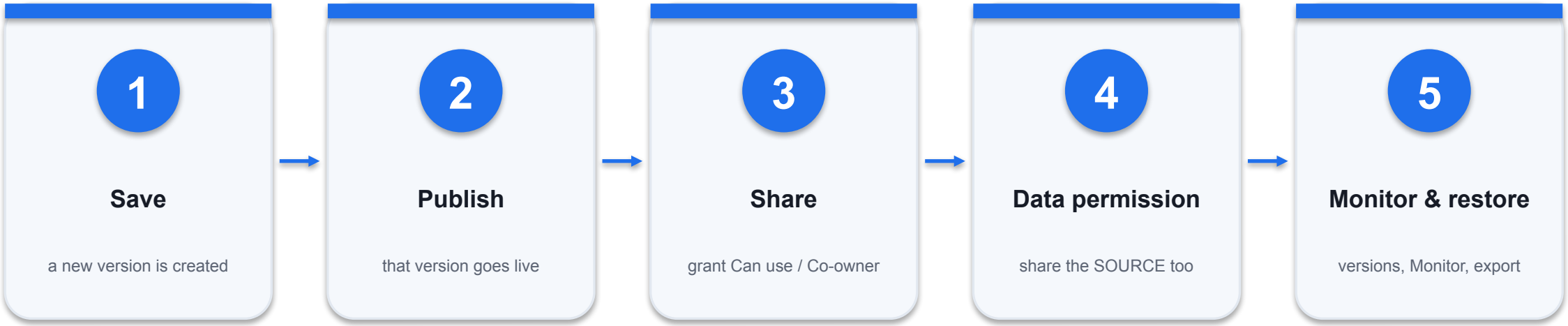
TOPIC 10

Publish, Share and Maintain a Canvas App

Publishing · sharing and permissions · versions · export · PCF components

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From Maker to Governed Asset

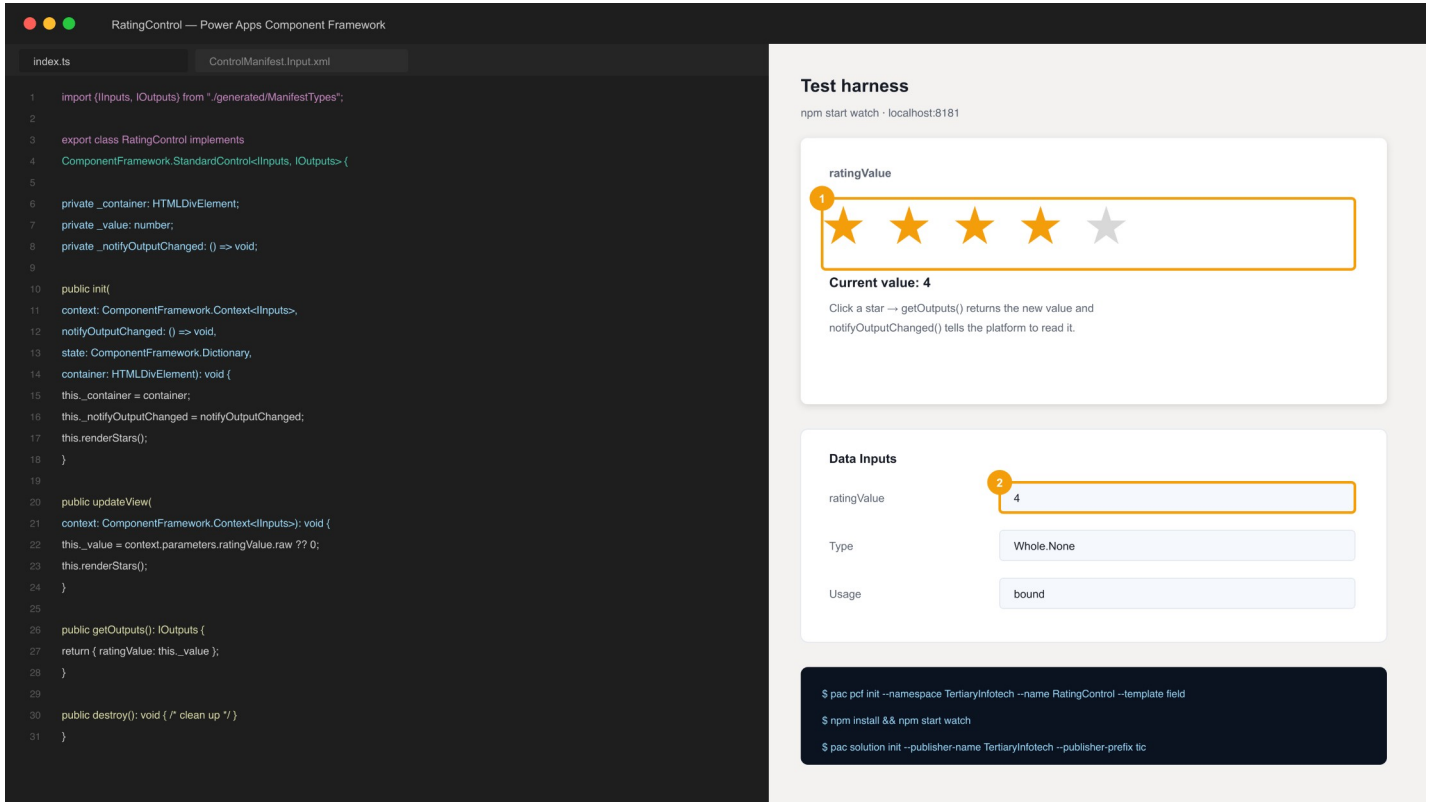


SHARING THE APP IS NOT SHARING THE DATA

A user with Can use on the app but no Dataverse security role opens the app and sees an error. This is the most common support call after a launch — and Lab 12 makes you reproduce it deliberately.

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Extending the Platform with a Code Component (PCF)



1. The rendered control in the local test harness — npm start watch, before any tenant is involved.

2. The bound property declared in the manifest. Its type must match the column you bind it to.

The Power Apps Component Framework is the API-level extension point: TypeScript in, a solution package out, usable in canvas and model-driven apps. Built in Lab 11.

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WHAT YOU'LL DO

Hands-On Labs — Create and Manage Canvas Apps with Power Apps

Labs 5–7

- Create a Canvas App from Excel Data
- Build the Dataverse Data Model
- Create a Canvas App over Dataverse

Labs 8–10

- Build and Theme the User Interface
- Work with External Data — SharePoint, Collections and Patch
- Write Data with Forms

Labs 11–12

- Extend the App with a Code Component (PCF)
- Publish, Share and Maintain the App

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LU2 · HANDS-ON

Create a Canvas App from Excel Data

LAB 5

A3 — Utilise a connector as middleware to integrate Excel data into an application.

Upload an Excel workbook to OneDrive for Business, connect Power Apps to its table through the Excel Online (Business) connector, and generate a three-screen canvas app from that data.

YOU'LL BUILD

A three-screen canvas app (browse, detail, edit) over an Excel table in OneDrive.

TOOLCHAIN

Power Apps, Excel Online (Business), OneDrive for Business

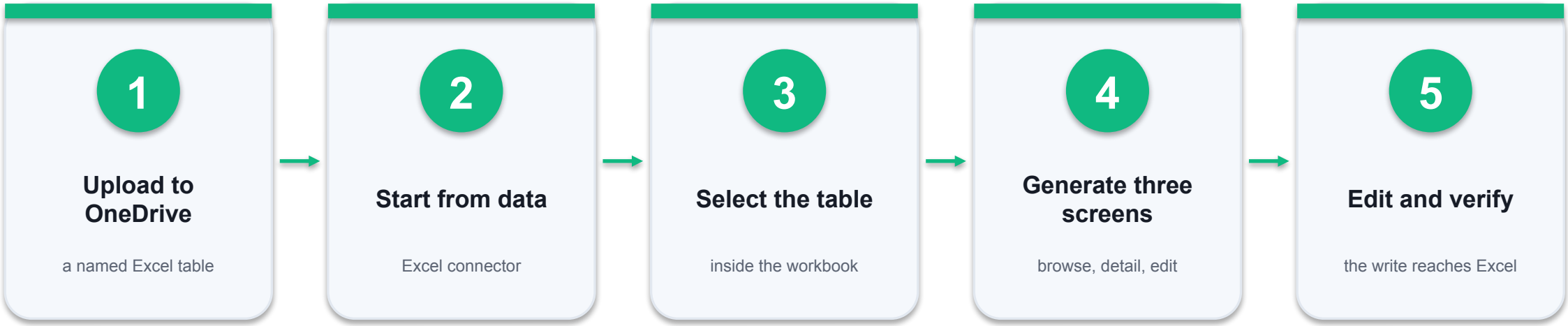
DONE WHEN

The app previews with data from Excel, and an edit made in the app appears in the Excel workbook in OneDrive.

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LAB 5 · PROCESS MAP

How Lab 5 Runs



THE POINT

A3 — Utilise a connector as middleware to integrate Excel data into an application.

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LAB 5 · VERIFY

Create a Canvas App from Excel Data

✅ Expected result

The app previews with data from Excel, and an edit made in the app appears in the Excel workbook in OneDrive.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-05/README.md.

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LU2 · HANDS-ON

Build the Dataverse Data Model

LAB 6

A1/A3 — Design tables, columns and a relationship as the integration point for apps and flows.

Create a Booking Request table in Dataverse with typed columns, then create a second table and join them with a lookup column so one row can reference another.

YOU'LL BUILD

Two related Dataverse tables with a lookup relationship and seeded rows.

TOOLCHAIN

Power Apps, Microsoft Dataverse

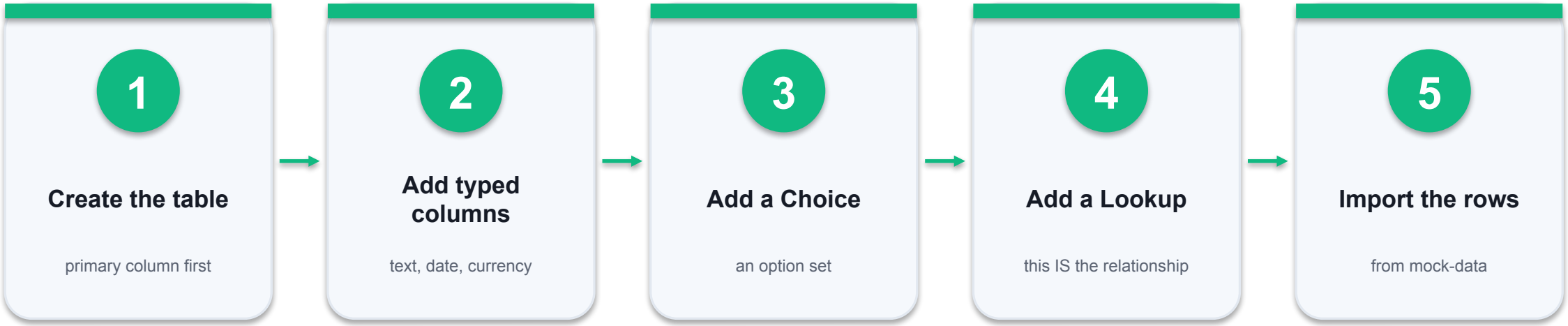
DONE WHEN

Booking Request shows the imported rows, the Status choice renders as a dropdown, and the Kennel lookup resolves to a Kennel row.

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LAB 6 · PROCESS MAP

How Lab 6 Runs



THE POINT

A1/A3 — Design tables, columns and a relationship as the integration point for apps and flows.

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LAB 6 · VERIFY

Build the Dataverse Data Model

✅ Expected result

Booking Request shows the imported rows, the Status choice renders as a dropdown, and the Kennel lookup resolves to a Kennel row.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-06/README.md.

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LU2 · HANDS-ON

Create a Canvas App over Dataverse

LAB 7

A3 — Bind a hand-built canvas app to Dataverse and format the data it displays.

Build a canvas app from blank, add Dataverse as a data source, lay out the main screen and add a gallery bound to the Booking Request table with formatted fields.

YOU'LL BUILD

A canvas app with a formatted, sorted gallery over the Booking Request table.

TOOLCHAIN

Power Apps, Microsoft Dataverse, Power Fx

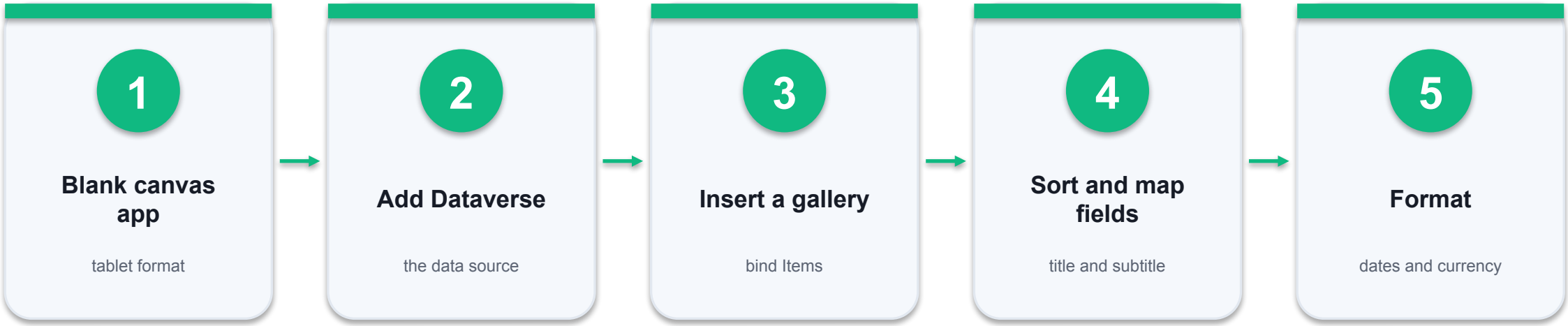
DONE WHEN

The gallery lists the Dataverse rows sorted by drop-off date, with dates and currency correctly formatted.

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LAB 7 · PROCESS MAP

How Lab 7 Runs



THE POINT

A3 — Bind a hand-built canvas app to Dataverse and format the data it displays.

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LAB 7 · VERIFY

Create a Canvas App over Dataverse

✅ Expected result

The gallery lists the Dataverse rows sorted by drop-off date, with dates and currency correctly formatted.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-07/README.md.

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LU2 · HANDS-ON

Build and Theme the User Interface

LAB 8

A3/A6 — Build a consistent, branded UI and verify it renders correctly across screens.

Apply a theme to the app, brand individual controls, and personalise the screen with the signed-in user's details from the Office 365 Users connector.

YOU'LL BUILD

A themed, branded canvas app that greets the signed-in user by name.

TOOLCHAIN

Power Apps, Power Fx, Office 365 Users

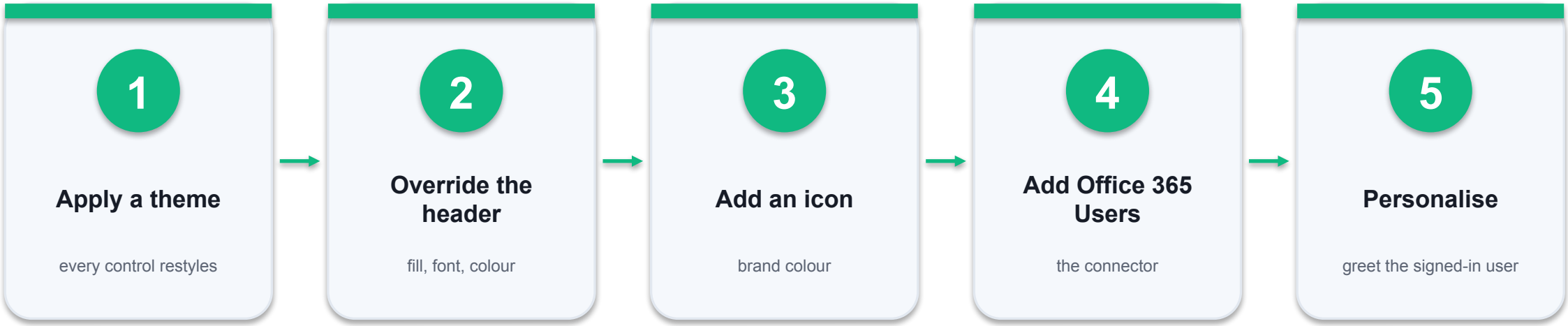
DONE WHEN

The app opens themed and branded, and the header greets the signed-in user by name with their profile photo.

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LAB 8 · PROCESS MAP

How Lab 8 Runs



THE POINT

A3/A6 — Build a consistent, branded UI and verify it renders correctly across screens.

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LAB 8 · VERIFY

Build and Theme the User Interface

✅ Expected result

The app opens themed and branded, and the header greets the signed-in user by name with their profile photo.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-08/README.md.

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LU2 · HANDS-ON

Work with External Data — SharePoint, Collections and Patch

LAB 9

A3/A4 — Integrate a second data source and write to it through API-level functions.

Create a SharePoint list, add it to the app as a second data source, load data into an in-memory collection, and use Patch to write status changes back to the list.

YOU'LL BUILD

A canvas app reading and writing both Dataverse and a SharePoint list, using a collection and Patch.

TOOLCHAIN

Power Apps, SharePoint Online, Power Fx, Office 365 Users

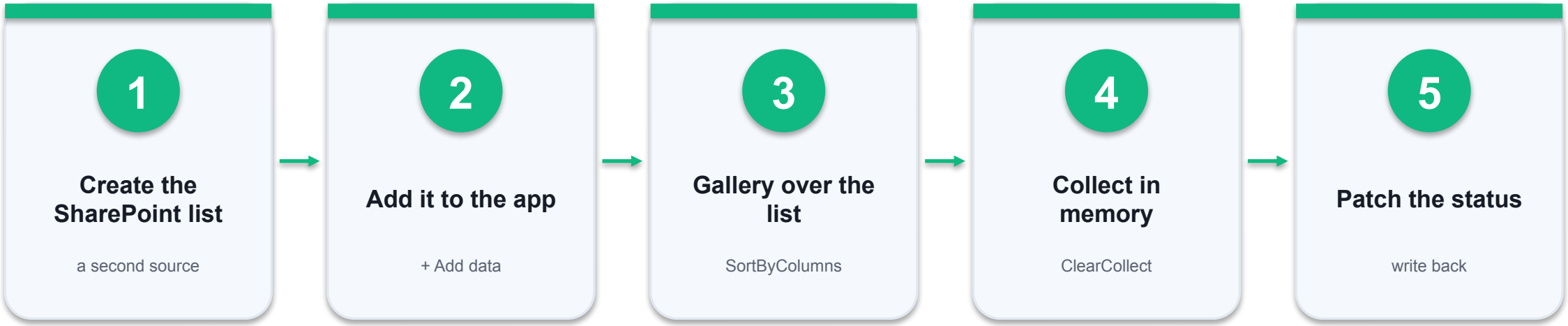
DONE WHEN

The SharePoint list shows the status values changed by the Confirm and Decline buttons, and colBookings appears under Variables > Collections.

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LAB 9 · PROCESS MAP

How Lab 9 Runs



THE POINT

A3/A4 — Integrate a second data source and write to it through API-level functions.

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LAB 9 · VERIFY

Work with External Data — SharePoint, Collections and Patch

✅ Expected result

The SharePoint list shows the status values changed by the Confirm and Decline buttons, and colBookings appears under Variables > Collections.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-09/README.md.

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LU2 · HANDS-ON

Write Data with Forms

LAB 10

A3/A5 — Use Form controls to display, edit and create records, and verify each write.

Add detail and edit screens, wire navigation between them, then use Display and Edit forms with SubmitForm and NewForm to read, update and create Dataverse records.

YOU'LL BUILD

A three-screen app with working display, edit, create and delete behaviour over Dataverse.

TOOLCHAIN

Power Apps, Microsoft Dataverse, Power Fx

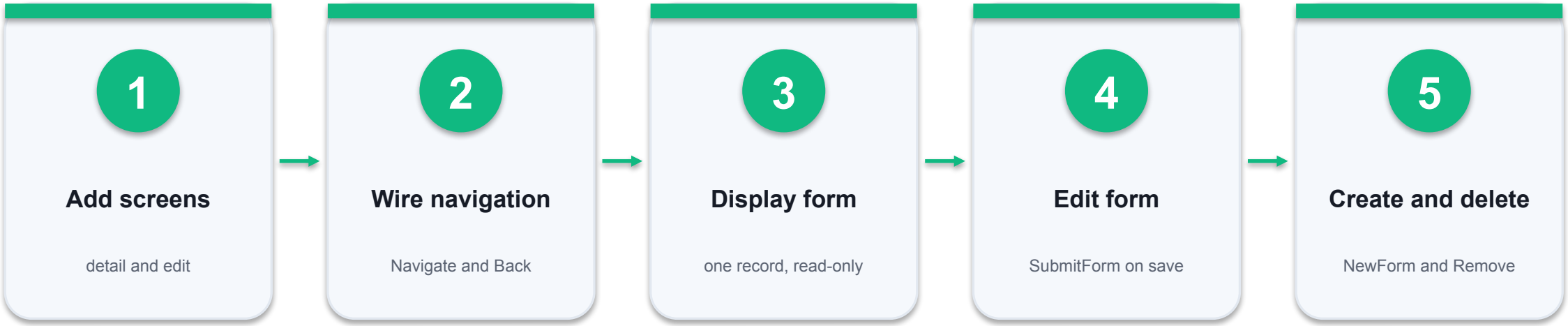
DONE WHEN

A record created in the app appears in Dataverse, an edit updates the same row, and Delete removes it — verified in the table's Data tab.

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LAB 10 · PROCESS MAP

How Lab 10 Runs



THE POINT

A3/A5 — Use Form controls to display, edit and create records, and verify each write.

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LAB 10 · VERIFY

Write Data with Forms

✅ Expected result

A record created in the app appears in Dataverse, an edit updates the same row, and Delete removes it — verified in the table's Data tab.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-10/README.md.

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LU2 · HANDS-ON

Extend the App with a Code Component (PCF)

LAB 11

A4/A8 — Support API-level integration by building a custom control and modifying the platform's UI layer.

Use the Power Apps Component Framework to build, test and package a custom code component, then import it into the environment and place it on a canvas app screen.

YOU'LL BUILD

A packaged PCF code component imported as a solution and used inside the canvas app.

TOOLCHAIN

Power Platform CLI, Node.js, TypeScript, Power Apps Component Framework, Dataverse solutions

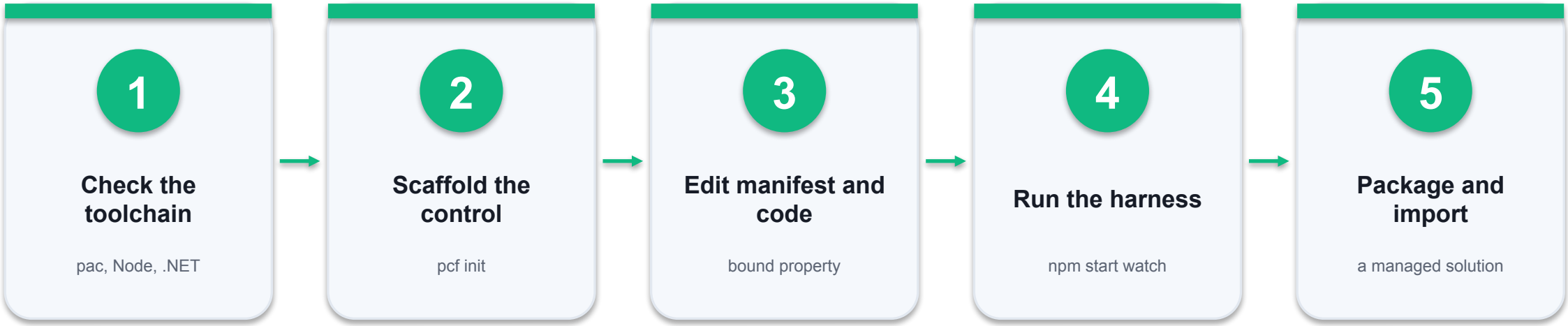
DONE WHEN

npm start watch renders the control in the harness, the solution imports without error, and the component can be inserted onto a canvas app screen.

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LAB 11 · PROCESS MAP

How Lab 11 Runs



THE POINT

A4/A8 — Support API-level integration by building a custom control and modifying the platform's UI layer.

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LAB 11 · VERIFY

Extend the App with a Code Component (PCF)

✅ Expected result

npm start watch renders the control in the harness, the solution imports without error, and the component can be inserted onto a canvas app screen.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-11/README.md.

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LU2 · HANDS-ON

Publish, Share and Maintain the App

LAB 12

A6/A8 — Verify the app functions for other users and manage its versions and permissions.

Publish the canvas app, share it with colleagues at the right permission level, review and restore versions, and export the app package for another environment.

YOU'LL BUILD

A published, shared and exported canvas app with a documented version history.

TOOLCHAIN

Power Apps, Microsoft Entra ID, Dataverse

DONE WHEN

The published app opens from its own URL for a second signed-in user, the version history shows a restore, and the exported .zip downloads.

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LAB 12 · PROCESS MAP

How Lab 12 Runs



THE POINT

A6/A8 — Verify the app functions for other users and manage its versions and permissions.

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LAB 12 · VERIFY

Publish, Share and Maintain the App

✅ Expected result

The published app opens from its own URL for a second signed-in user, the version history shows a restore, and the exported .zip downloads.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-12/README.md.

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LEARNING UNIT RECAP · YOU CAN NOW

Recap — Create and Manage Canvas Apps with Power Apps (1/2)

1

Lab 5 — Create a Canvas App from Excel Data

A3 — Utilise a connector as middleware to integrate Excel data into an application.

2

Lab 6 — Build the Dataverse Data Model

A1/A3 — Design tables, columns and a relationship as the integration point for apps and flows.

3

Lab 7 — Create a Canvas App over Dataverse

A3 — Bind a hand-built canvas app to Dataverse and format the data it displays.

4

Lab 8 — Build and Theme the User Interface

A3/A6 — Build a consistent, branded UI and verify it renders correctly across screens.

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LEARNING UNIT RECAP · YOU CAN NOW

Recap — Create and Manage Canvas Apps with Power Apps (2/2)

1

Lab 9 — Work with External Data — SharePoint, Collections and Patch

A3/A4 — Integrate a second data source and write to it through API-level functions.

2

Lab 10 — Write Data with Forms

A3/A5 — Use Form controls to display, edit and create records, and verify each write.

3

Lab 11 — Extend the App with a Code Component (PCF)

A4/A8 — Support API-level integration by building a custom control and modifying the platform's UI layer.

4

Lab 12 — Publish, Share and Maintain the App

A6/A8 — Verify the app functions for other users and manage its versions and permissions.

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3

LU3

Automated Processes with Power Automate and Copilot Studio Agents

Topics 11–14 · Labs 13–20

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11

TOPIC 11

Get Started with Power Automate

Flow anatomy · triggers · actions · dynamic content · run history

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TOPIC 11 · CONCEPT

Every Flow Is the Same Four Parts



LEARN IT ONCE

Every flow in Labs 13–20 is this shape. Only the trigger and the actions change. A flow has exactly ONE trigger — change it and you have a different flow, even if the actions are identical.

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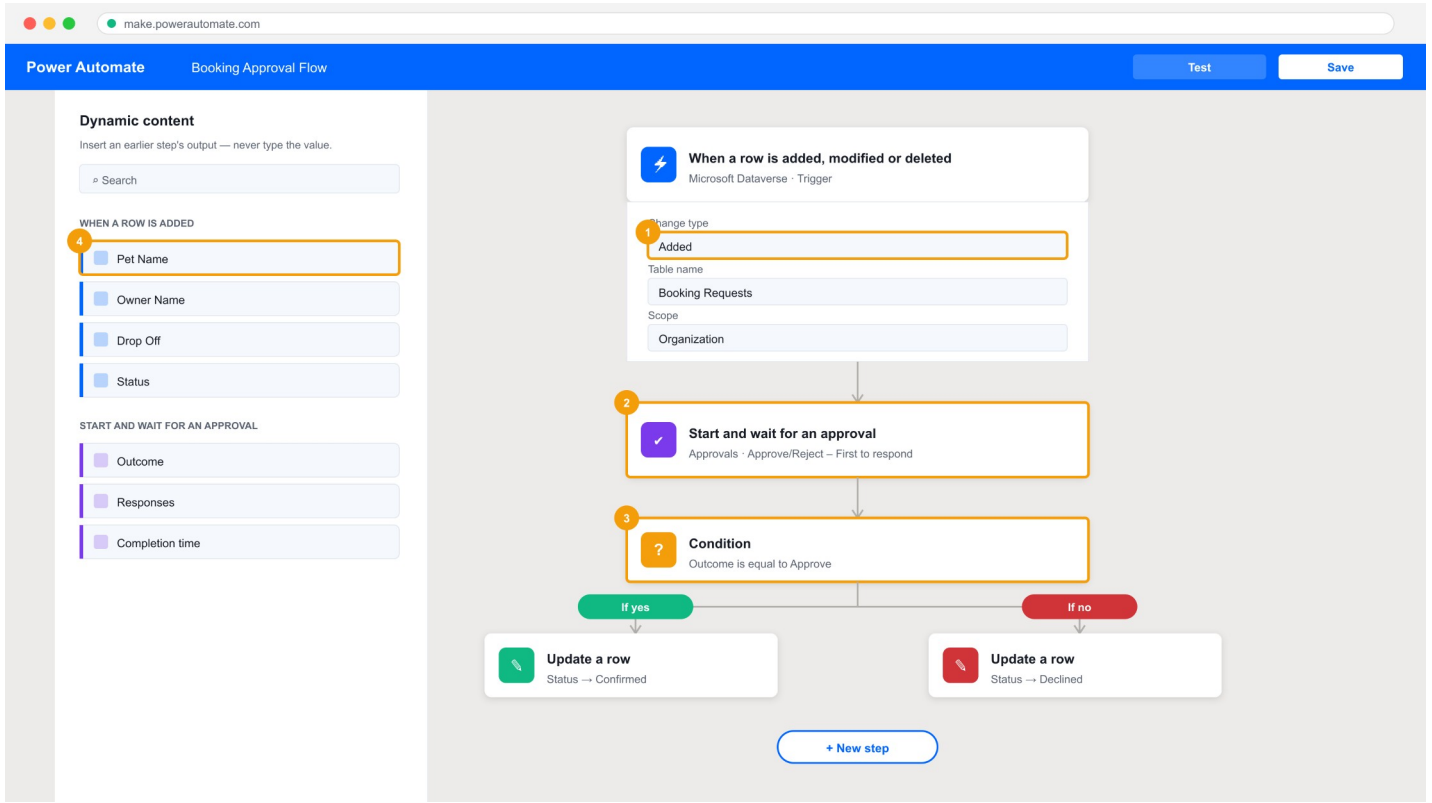
Four Trigger Families — Who Is Allowed to Start the Process

Family	What starts it	Example trigger	Used in
Manual	A person presses Run	Manually trigger a flow · button in Teams	Lab 15
Scheduled	A clock reaches a time	Recurrence — set the TIME ZONE	Lab 15
Automated	An event happens in a system	When a row is added · when an item is created	Labs 13, 14, 16
Request	Something calls in from outside	HTTP request received · when an agent calls the workflow	Labs 18, 19

WHEN IT MATTERS The trigger you choose is a statement about who or what may start this process. A Recurrence left on UTC runs at the wrong hour in Singapore — set the time zone explicitly.

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The Power Automate Designer



1. The trigger's own settings — which table, which change type, what scope.
2. An action. Each one runs after the one above it.
3. A Condition branches the run on a value produced earlier.
4. Dynamic content — insert an earlier step's OUTPUT as a token. Never type the value.

Applied in Labs 13–16. The left panel is the single most important habit in the course: insert, never type.

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TOPIC 11 · THE HABIT

A typed value is a constant that will be wrong tomorrow.

Dynamic content inserts the actual output of an earlier step at run time. Typing the same text produces a flow that works once, for one record, and silently sends the wrong data forever after.

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TOPIC 11 · VERIFICATION

Reading Run History — How You Test a Flow (TSC A5 / A6)

1

Open the run

My flows › the flow › Run history › the run you care about.

2

Read the trigger

What actually fired it, and what values it carried in.

3

Expand each action

Inputs = what it was given. Outputs = what it produced.

4

Find the first failure

Everything after it is a consequence, not a cause.

5

Read the error text

404 = not found. 403 = permission. 429 = throttled.

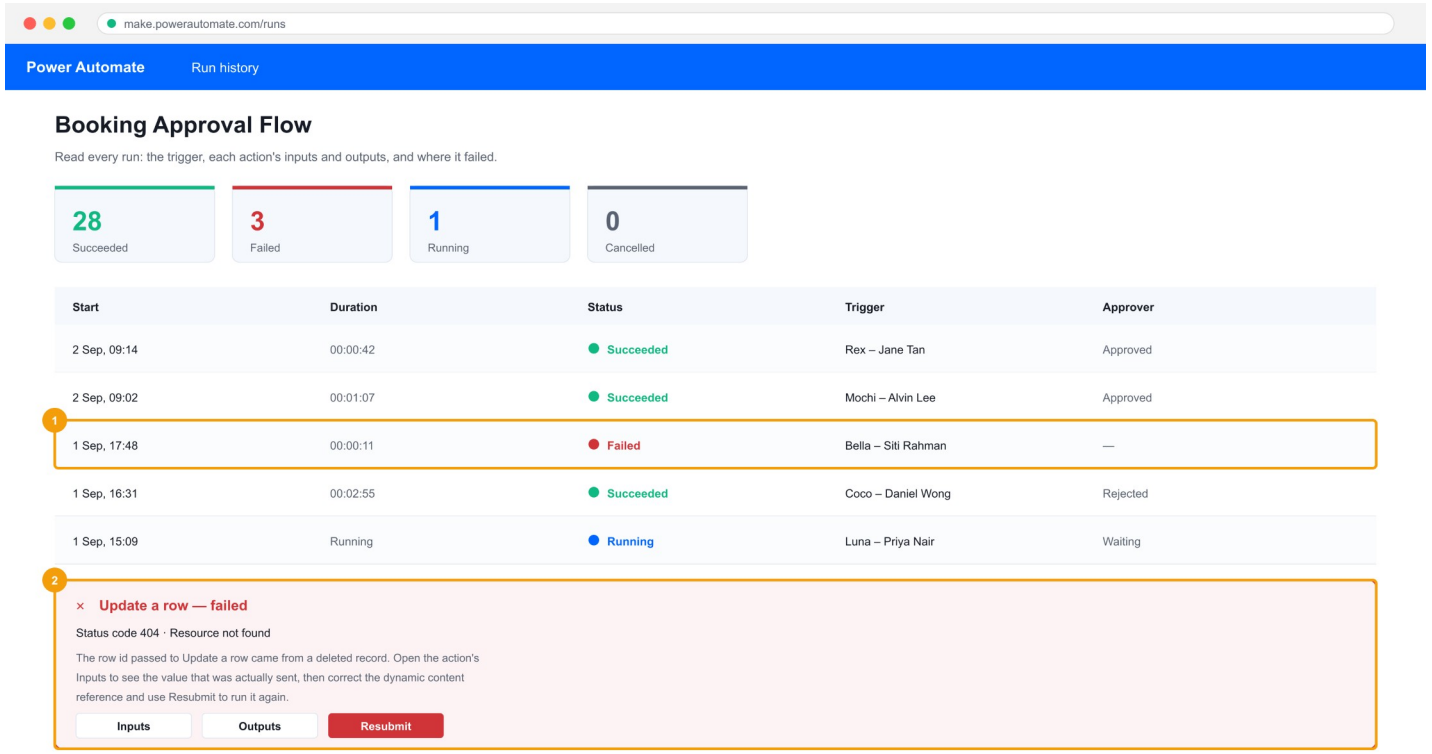
6

Fix, then Resubmit

Resubmit re-runs the SAME trigger data — no need to recreate the event.

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Run History Is the Evidence



1. A failed run. Status and duration are the first signal something is wrong.
2. The failed action, its error, and Resubmit — the fix-and-retest loop.

This screen is your TSC A5/A6 evidence: it shows the connection between disparate programs, and whether it functioned.

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12

TOPIC 12

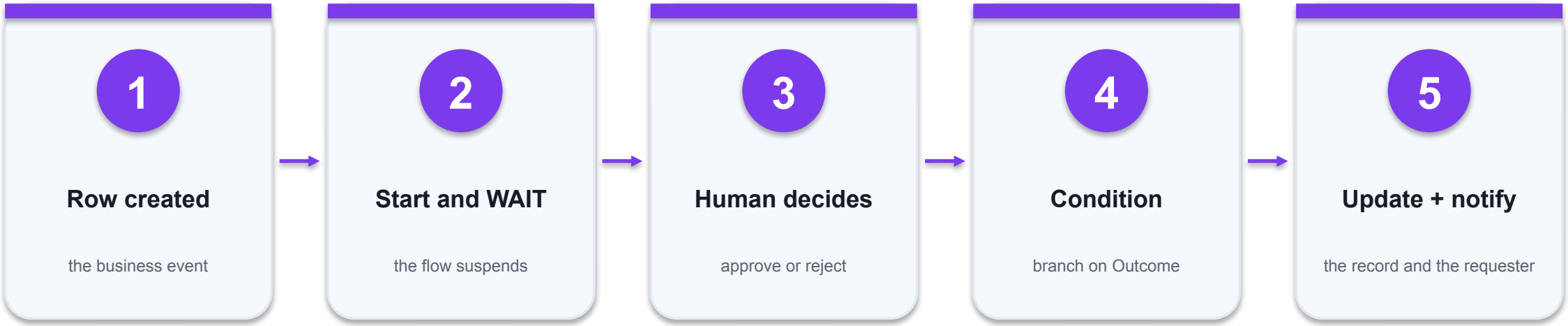
Build Approval Flows with Power Automate

Approvals · outcome branching · business process flows · human in the loop

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TOPIC 12 · CONCEPT

An Approval Pauses the Flow for a Person

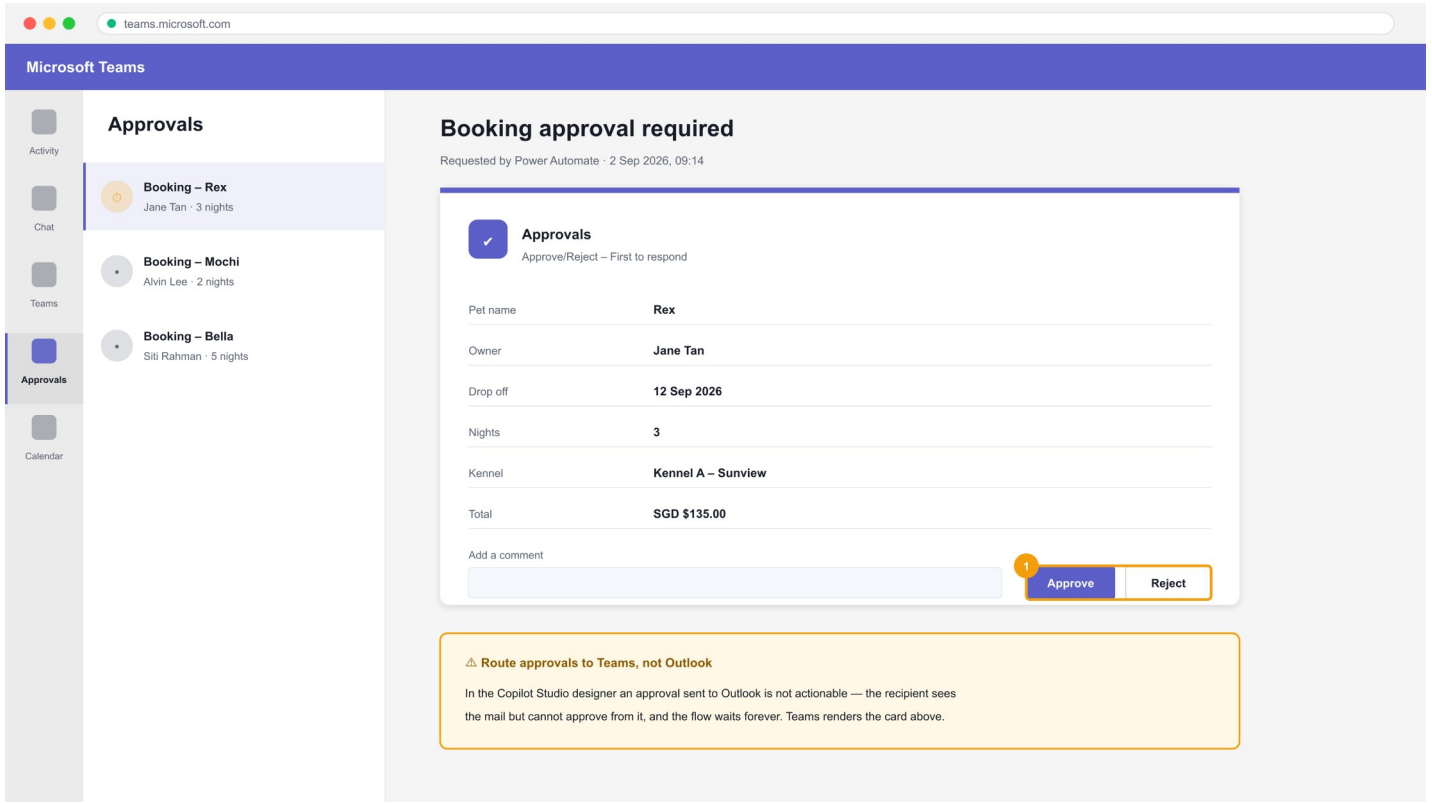


'START AND WAIT' IS THE POINT

The flow genuinely suspends — for minutes or days — and resumes on the human's answer. This is how you put a person where the consequence lands.

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The Approval Card — Route It to Teams



1. Approve and Reject are actionable IN the card. The flow resumes the moment one is pressed.

Verified on a live tenant: an approval sent to Outlook is not actionable in the Copilot Studio designer — the recipient can read it but cannot respond, and the flow waits forever. Send approvals to Teams.

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Approval Types — Choose by How the Decision Is Made

Type	Who must respond	Outcome returned	Use when
Approve/Reject – First to respond	Any ONE of the assignees	Approve or Reject	Speed matters; any manager may decide
Approve/Reject – Everyone must approve	ALL assignees	Approve only if all approve	Spend or risk needs unanimous sign-off
Custom Responses – Wait for one	Any one assignee	Your own option text	The decision is not binary
Custom Responses – Wait for all	All assignees	All the responses	Multi-party sign-off with reasons

WHEN IT MATTERS The Outcome value is a STRING you compare in a Condition — 'Approve', not 'Approved'. Getting this wrong sends every request down the reject branch, and the flow still reports Succeeded.

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TOPIC 13

Build Flows to Manage User Information

Trigger filters · user context · run-time input · calendar and location

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Trigger Filter or Condition — Where Should the Test Live?



NO — use a TRIGGER FILTER

Select columns + Filter rows on the trigger. The flow never starts, so it costs nothing and the run history stays readable.

YES, but branch — use a CONDITION

The flow runs, then chooses a path. Every evaluation is a real run you will see in the history.

Lab 16 measures this: the same edits before and after filtering, with the run counts compared. Filters use the SCHEMA name (statuscode), never the display name.

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Getting the Right Values Into a Flow

1

Trigger inputs

Ask the person at run time — Text, Number, Date, File, Email.

2

Office 365 Users

Get my profile, Get manager — resolve the requester's chain of command.

3

Dynamic content

Any earlier step's output, inserted as a token.

4

Expressions

utcNow(), formatDateTime(), addDays() — for values nothing produced.

5

Variables

Initialize once, Set or Append inside a loop.

6

Parse JSON

Turn a JSON string into fields — but see structured output in Topic 14.

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14

TOPIC 14

Power Automate, Copilot Studio Agents and Deep Integration

The new Copilot Studio · workflow vs agent · agent anatomy · tools · what is enforced

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TOPIC 14 · CONCEPT

Workflow or Agent — They Fail Differently

	WORKFLOW — Power Automate	AGENT — Copilot Studio
The path	You decide it in advance	It chooses at run time
Same input twice	Always the same output	May give a different answer
Capability	Only what you built	Can combine what it was given in new ways
How it fails	Loudly, at a named step	Quietly, with a confident wrong answer
How you test it	Check the result	Try to BREAK it
In this course	Labs 13–16	Labs 17–20

WHEN IT MATTERS Use a workflow where the rule is known. Use an agent where the language varies. Most real systems need both — and the seam between them is where the design work is.

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The New Copilot Studio — a Model Inside a Harness

1

The AI model — the brain

The reasoning engine. It understands language, spots patterns and decides what to do next. Chosen per agent from the Model dropdown — GPT-5.x or Claude.

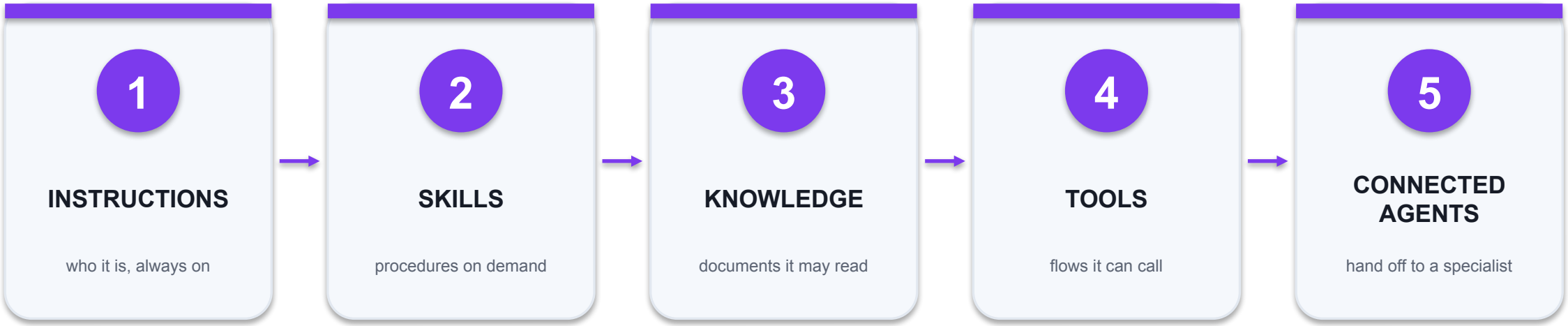
2

The harness — around it

The orchestration layer the model works inside: context, tools, governance, workflows and security. Model-agnostic — it turns reasoning into governed work.

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The Anatomy of an Agent — Five Parts



BUILT IN LABS 17–20

Lab 17 builds instructions and knowledge. Lab 18 adds a tool. Lab 19 puts an Agent node inside a flow. Lab 20 adds a connected agent and a human gate.

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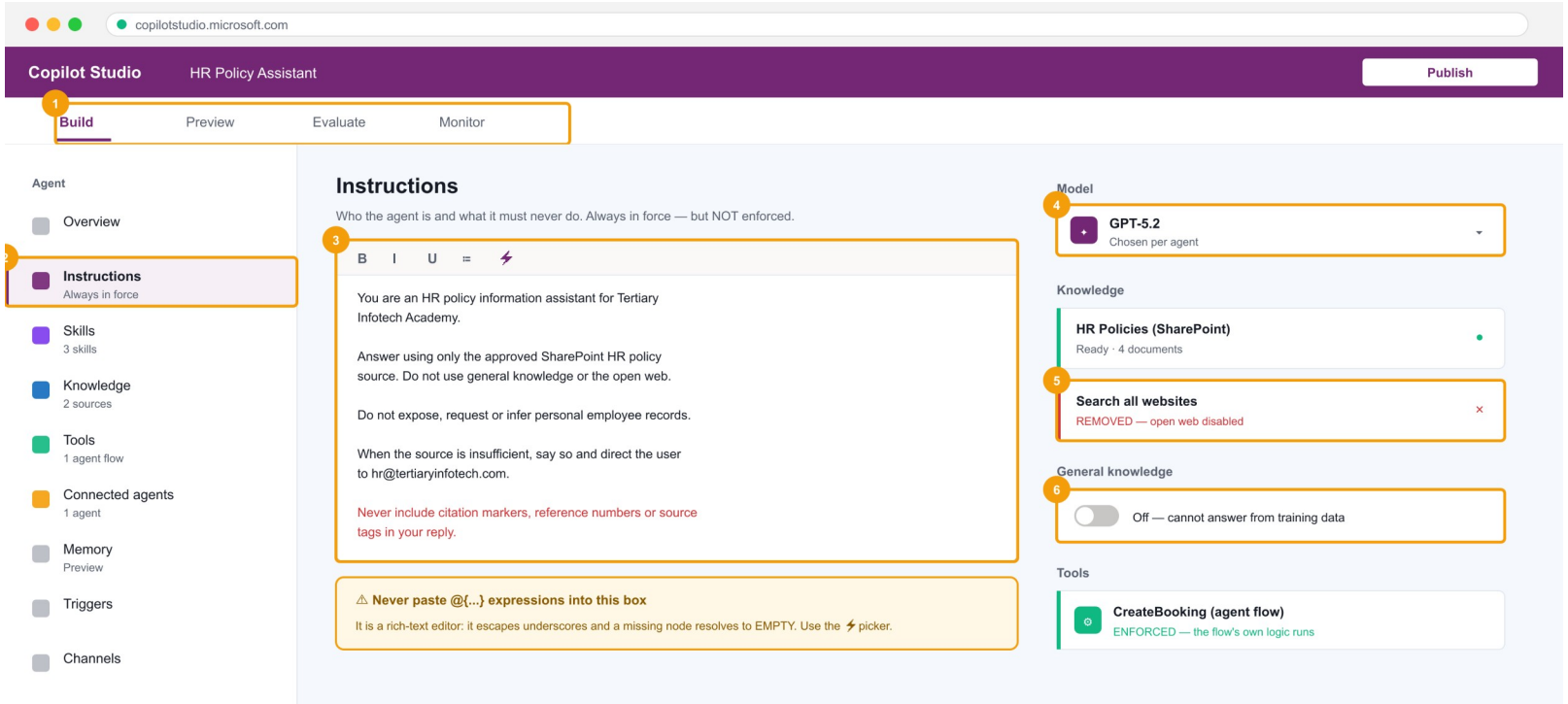
Which Parts of an Agent Hold — and Which Do Not

Part	What it is	Enforced?
Instructions	Who the agent is; always in force	NO — probabilistic
Skill	A named procedure for a matching topic	NO — the model decides it applies
Knowledge	Documents the agent may read	PARTLY — it cannot read what it lacks
Tool (agent flow)	A flow that acts outside the conversation	YES — the flow's own logic runs
Connected agent	A separate agent with its own knowledge	YES — the knowledge boundary is real

WHEN IT MATTERS A control the model cannot reach beats a rule you asked it to follow. So: a tool the agent does NOT have is a control — and the schema beats the prompt, because a field that exists will eventually be filled.

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The New Agent Designer — Everything on One Rail



1. Four tabs: Build, Preview, Evaluate, Monitor — author, probe, score, then watch it live.
2. Instructions, Skills, Knowledge, Tools, Connected agents and Memory, all on one rail.
3. The instructions box is a RICH-TEXT editor. Never paste @{...} into it.
4. The model is chosen per agent; the harness stays the same.
5. 'Search all websites' is ON by default. Remove it, or open-web facts reach your user.
6. Turn general knowledge OFF, or the model answers from training data and you cannot tell which answers those were.

Grounding is not only about giving the agent facts. It is about taking away every other source of them. Built in Lab 17.

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Writing Instructions That Constrain

1

1 · Identity

"You are an HR policy information assistant." One role, one line.

2

2 · Source rule

"Answer using only the approved SharePoint HR policy source."

3

3 · Refusals

"Do not expose, request or infer personal employee records." Prohibitions, not preferences.

4

4 · Escalation

"When the source is insufficient, say so and direct the user to HR."

5

5 · No citation markers

Grounding appends [doc:turn1doc11] and [1] tags the model did not write. Ban them explicitly.

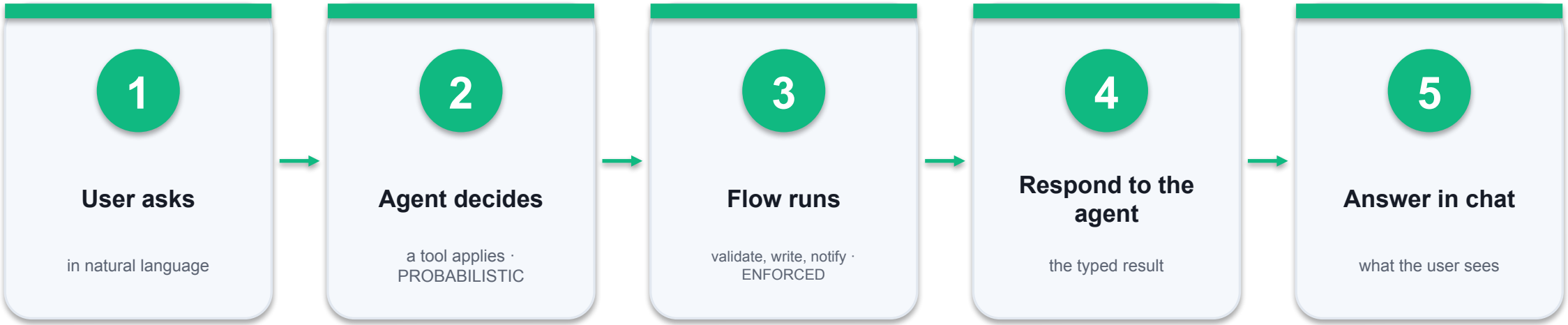
6

6 · Build with the ⚡ picker

The editor escapes underscores, and a missing node resolves to EMPTY rather than erroring.

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Tools — Where the Agent Stops and the Flow Starts

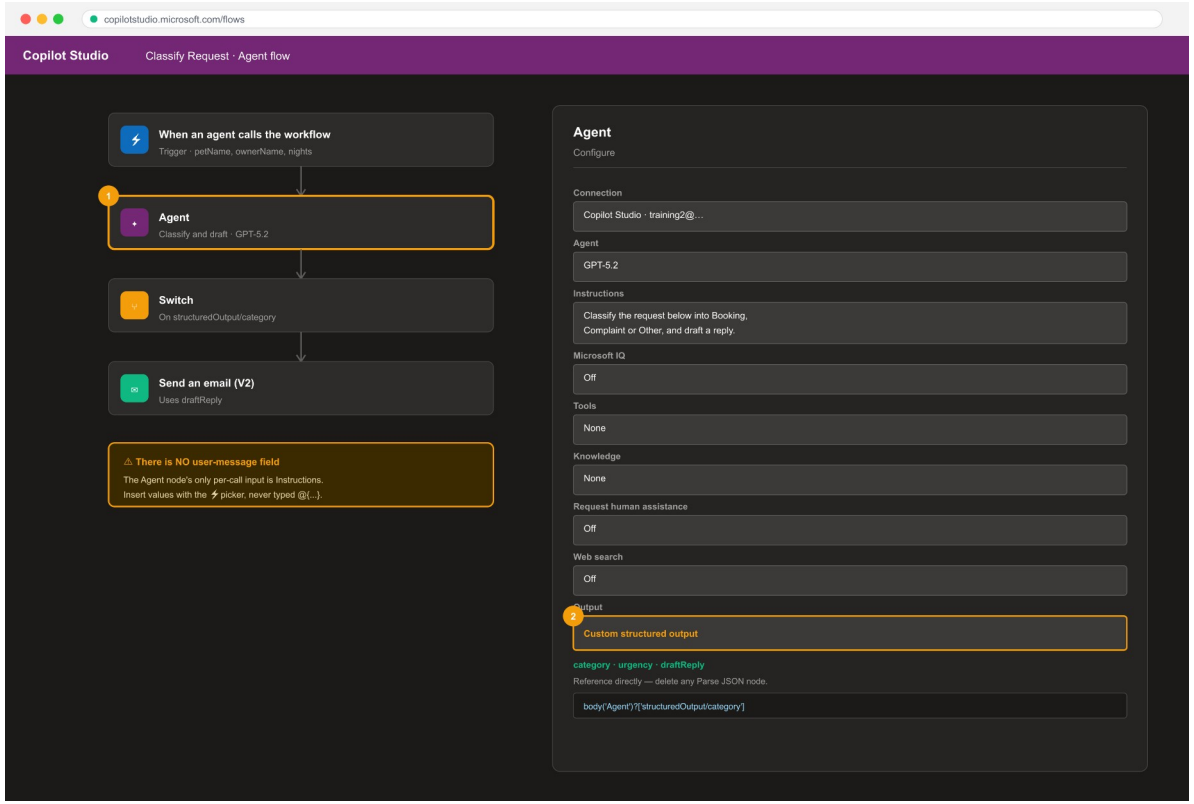


THE SEAM IS THE DESIGN

The agent may call the wrong tool, or the right one with wrong values. Whatever it believed, the flow's own validation still runs. Put the consequence inside the flow — Lab 18.

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The Agent Node Inside a Flow — and Structured Output



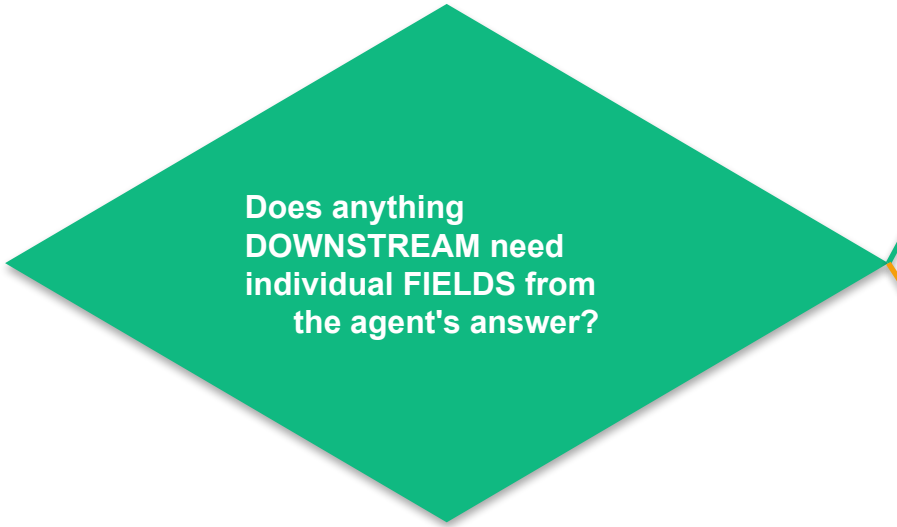
1. The Agent node. Its Configure tab has NO user-message field — Instructions is the only per-call input.

2. Output › Custom structured output. Define the schema, then reference fields directly and DELETE any Parse JSON node.

With Text response you are asking the model for JSON and hoping. With Custom structured output the fields are typed — body('Agent')?['structuredOutput/category']. Built in Lab 19.

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Structured Output or Parse JSON?



YES — Custom structured output

Define the schema on the Agent node. Reference fields with the slash syntax. Delete the Parse JSON node entirely.

NO — Text response is fine

The answer goes straight into an email body or a chat reply. Nothing branches on it.

With Text response the model will eventually wrap its answer in ```json fences or open with 'Here is the JSON', and Parse JSON fails with "Error parsing NaN value. Path '', line 1, position 1" — position 1 means the first character was not {.

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WHAT YOU'LL DO

Hands-On Labs — Create and Manage Automated Processes with Power Automate and Copilot Studio Agents

Labs 13–15

- Create Cloud Flows — Template, Copilot and Blank
- Build an Approval Flow with a Human in the Loop
- Button Flows and Scheduled Flows

Labs 16–18

- Trigger Filters and Cross-Source Integration
- Build Your First Copilot Studio Agent
- Give the Agent a Tool — Calling a Workflow from an Agent

Labs 19–20

- Call an Agent from a Workflow with Structured Output
- Multi-Agent Handoff and a Human Review Gate

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LU3 · HANDS-ON

Create Cloud Flows — Template, Copilot and Blank

LAB 13

A3/A5 — Build flows three ways and verify each run against its trigger and actions.

Create one flow from a template, one with Copilot and one from blank, so the trigger-actions-outputs anatomy is visible from three directions, then read the run history of each.

YOU'LL BUILD

Three working cloud flows plus a read of their run history.

TOOLCHAIN

Power Automate, Outlook, SharePoint, Copilot for Power Automate

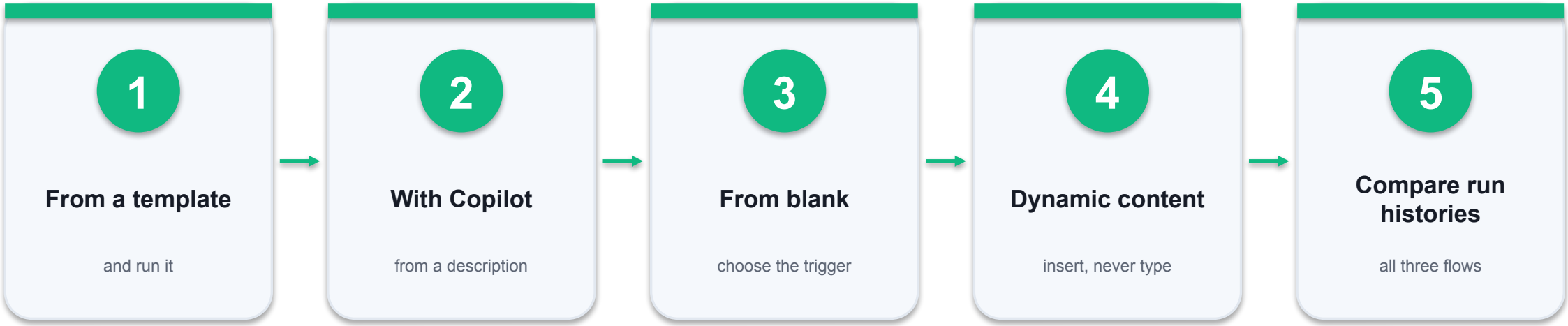
DONE WHEN

All three flows show a Succeeded run, and you can name the trigger and each action's output that supplied a value in the email.

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LAB 13 · PROCESS MAP

How Lab 13 Runs



THE POINT

A3/A5 — Build flows three ways and verify each run against its trigger and actions.

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LAB 13 · VERIFY

Create Cloud Flows — Template, Copilot and Blank

✅ Expected result

All three flows show a Succeeded run, and you can name the trigger and each action's output that supplied a value in the email.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-13/README.md.

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LU3 · HANDS-ON

Build an Approval Flow with a Human in the Loop

LAB 14

A3/A5 — Pause an automated process for a human decision and branch on the outcome.

Build a flow that starts when a booking request row is created, sends an approval to a manager, waits for the response, and updates the record's status based on the outcome.

YOU'LL BUILD

An approval flow that writes Confirmed or Declined back to Dataverse based on a human decision.

TOOLCHAIN

Power Automate, Approvals, Dataverse, Outlook, Microsoft Teams

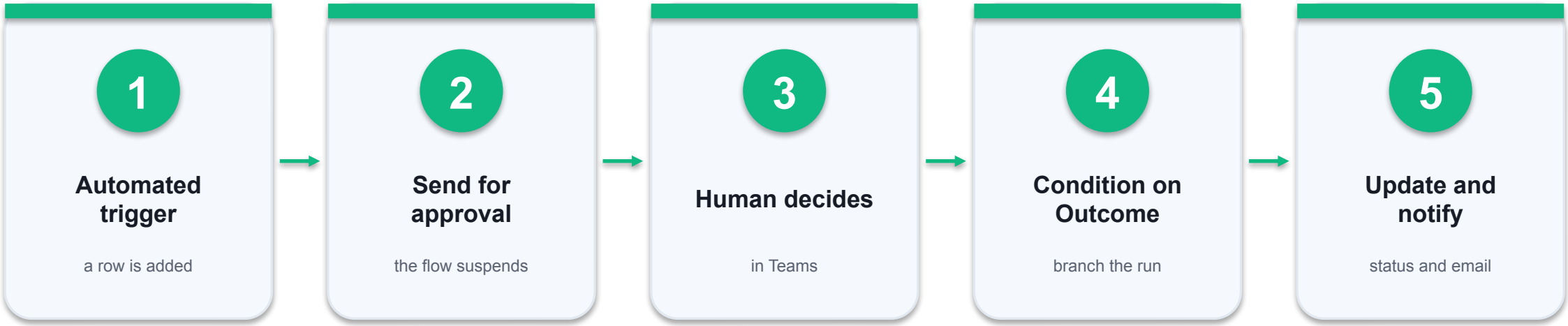
DONE WHEN

Approving the request sets the Dataverse row's Status to Confirmed and the requester receives the notification email; rejecting sets Declined.

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LAB 14 · PROCESS MAP

How Lab 14 Runs



THE POINT

A3/A5 — Pause an automated process for a human decision and branch on the outcome.

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LAB 14 · VERIFY

Build an Approval Flow with a Human in the Loop

✅ Expected result

Approving the request sets the Dataverse row's Status to Confirmed and the requester receives the notification email; rejecting sets Declined.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-14/README.md.

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LU3 · HANDS-ON

Button Flows and Scheduled Flows

LAB 15

A3 — Use manual and scheduled triggers, and collect user input at run time.

Build an instant (button) flow that prompts the user for input and creates a record, then build a scheduled flow that queries rows on a recurrence and emails an HTML summary table.

YOU'LL BUILD

An instant flow with user input and a recurring digest flow that emails a formatted table.

TOOLCHAIN

Power Automate, Dataverse, Outlook

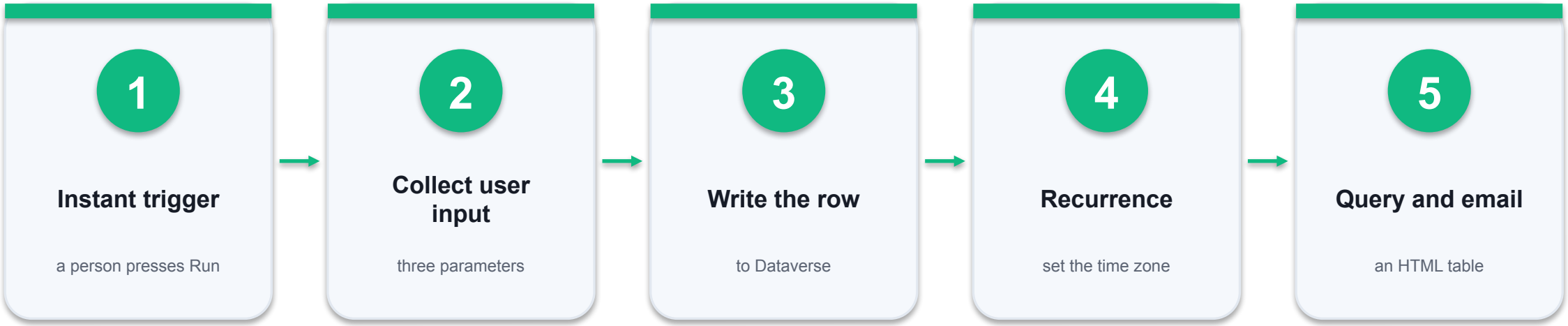
DONE WHEN

The button flow creates a Dataverse row from the values you typed, and the scheduled flow's test run emails a formatted HTML table of pending bookings.

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LAB 15 · PROCESS MAP

How Lab 15 Runs



THE POINT

A3 — Use manual and scheduled triggers, and collect user input at run time.

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LAB 15 · VERIFY

Button Flows and Scheduled Flows

✅ Expected result

The button flow creates a Dataverse row from the values you typed, and the scheduled flow's test run emails a formatted HTML table of pending bookings.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-15/README.md.

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LU3 · HANDS-ON

Trigger Filters and Cross-Source Integration

LAB 16

A5/A7 — Stop unnecessary runs with trigger filters and highlight performance issues.

Add column and row filters to an automated trigger so the flow only runs when it must, then integrate SharePoint, Dataverse and Excel in one flow and compare run counts before and after filtering.

YOU'LL BUILD

A filtered automated flow plus a run-count comparison showing the performance improvement.

TOOLCHAIN

Power Automate, Dataverse, SharePoint, Excel Online (Business)

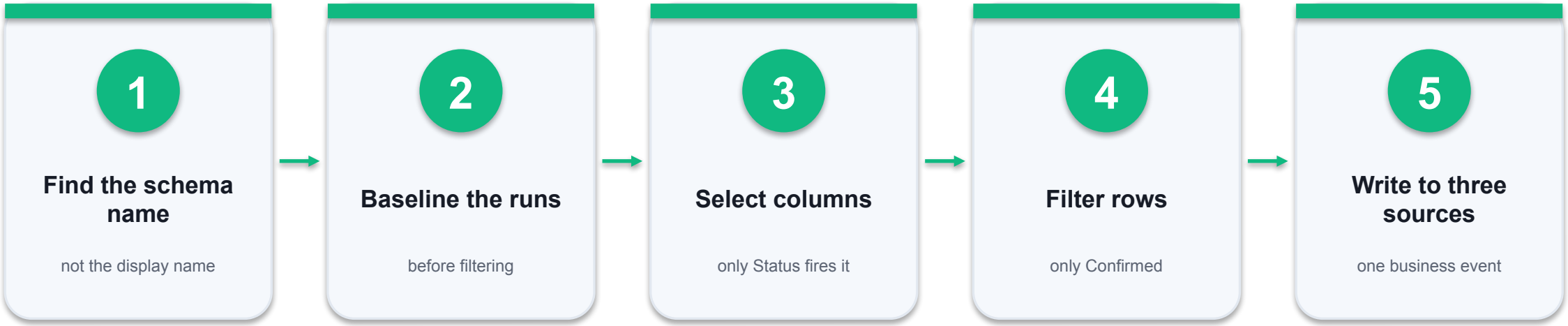
DONE WHEN

After filtering, edits to non-Status columns produce no runs, and a single Confirmed change writes to all three data sources in one Succeeded run.

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LAB 16 · PROCESS MAP

How Lab 16 Runs



THE POINT

A5/A7 — Stop unnecessary runs with trigger filters and highlight performance issues.

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LAB 16 · VERIFY

Trigger Filters and Cross-Source Integration

✅ Expected result

After filtering, edits to non-Status columns produce no runs, and a single Confirmed change writes to all three data sources in one Succeeded run.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-16/README.md.

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LU3 · HANDS-ON

Build Your First Copilot Studio Agent

LAB 17

A1/A3 — Build a conversational agent with instructions and knowledge as a new integration surface.

In the new Copilot Studio agent designer, create an agent, choose its model, write constraining instructions, attach a SharePoint knowledge source and remove every other source of facts, then probe it for hallucination.

YOU'LL BUILD

A grounded Copilot Studio agent that answers only from your documents and refuses otherwise.

TOOLCHAIN

Copilot Studio (new agent designer), SharePoint, Dataverse

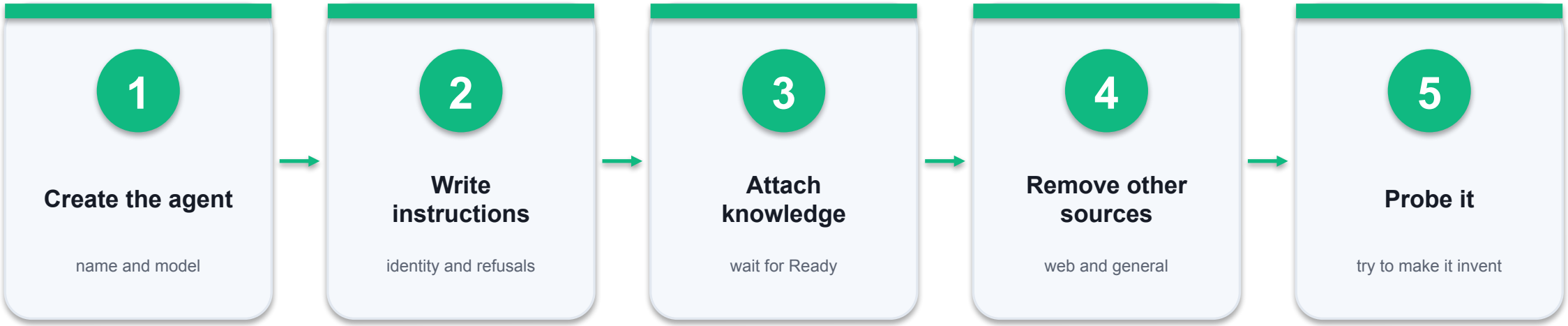
DONE WHEN

The agent answers the in-scope questions from the documents and explicitly declines the out-of-scope ones instead of inventing an answer.

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LAB 17 · PROCESS MAP

How Lab 17 Runs



THE POINT

A1/A3 — Build a conversational agent with instructions and knowledge as a new integration surface.

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LAB 17 · VERIFY

Build Your First Copilot Studio Agent

✅ Expected result

The agent answers the in-scope questions from the documents and explicitly declines the out-of-scope ones instead of inventing an answer.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-17/README.md.

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LU3 · HANDS-ON

Give the Agent a Tool — Calling a Workflow from an Agent

LAB 18

A4 — Expose a deterministic flow to an agent as a callable tool with typed inputs and outputs.

Build an agent flow that validates and writes a booking, then register it as a tool on the agent so the conversation can act on the outside world while the enforced logic stays in the flow.

YOU'LL BUILD

An agent that calls a Power Automate flow as a tool and returns the flow's result in chat.

TOOLCHAIN

Copilot Studio agent flows, Power Automate, Dataverse, Microsoft Teams

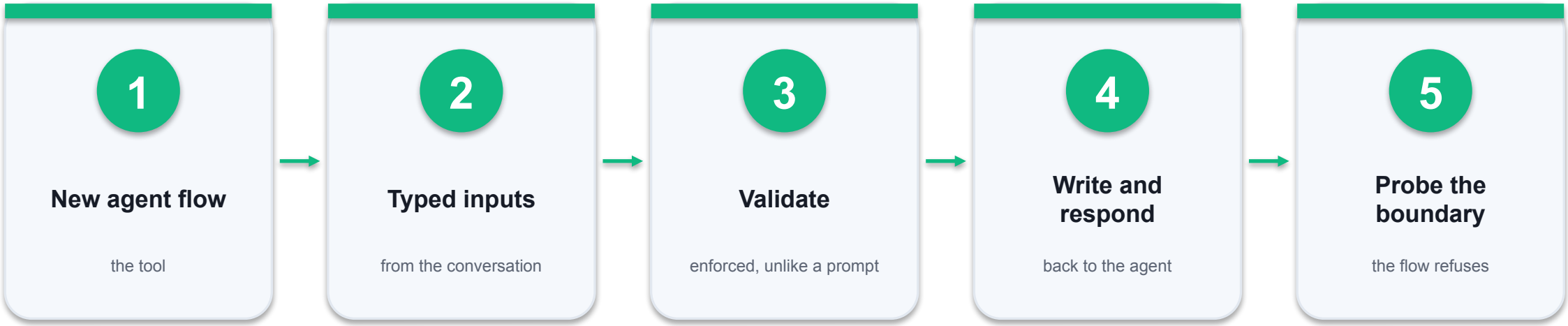
DONE WHEN

A valid request creates a Dataverse row and returns the confirmation in chat; the 60-night request is refused by the flow's own Condition.

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LAB 18 · PROCESS MAP

How Lab 18 Runs



THE POINT

A4 — Expose a deterministic flow to an agent as a callable tool with typed inputs and outputs.

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LAB 18 · VERIFY

Give the Agent a Tool — Calling a Workflow from an Agent

✅ Expected result

A valid request creates a Dataverse row and returns the confirmation in chat; the 60-night request is refused by the flow's own Condition.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-18/README.md.

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LU3 · HANDS-ON

Call an Agent from a Workflow with Structured Output

LAB 19

A4/A5 — Use an Agent node inside a flow and consume its typed fields downstream.

Build a Power Automate flow containing an Agent node that classifies an incoming request, configure Custom structured output so the result is typed JSON, and branch the flow on the returned fields.

YOU'LL BUILD

A flow whose Agent node returns structured JSON that the following actions branch on.

TOOLCHAIN

Copilot Studio workflow designer, Agent node, Dataverse, Outlook

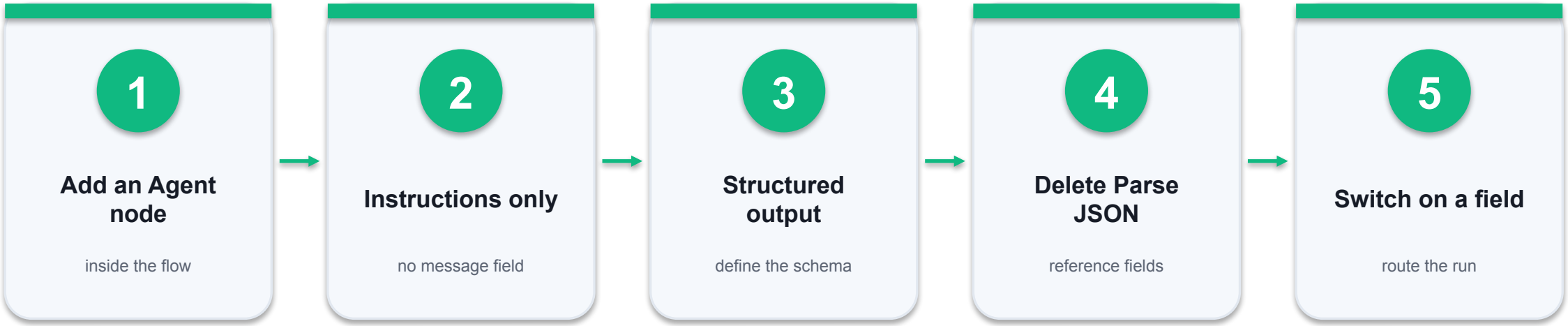
DONE WHEN

The Agent node's Run Details shows the three schema fields populated, and the Switch routes the run to the case matching the returned category.

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LAB 19 · PROCESS MAP

How Lab 19 Runs



THE POINT

A4/A5 — Use an Agent node inside a flow and consume its typed fields downstream.

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LAB 19 · VERIFY

Call an Agent from a Workflow with Structured Output

✅ Expected result

The Agent node's Run Details shows the three schema fields populated, and the Switch routes the run to the case matching the returned category.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-19/README.md.

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LU3 · HANDS-ON

Multi-Agent Handoff and a Human Review Gate

LAB 20

A5/A7 — Compose connected agents and place a human where the consequence lands.

Connect a second specialist agent so one conversation can hand off to another with its own knowledge, then add a Request human assistance gate so a consequential action cannot proceed without approval.

YOU'LL BUILD

A two-agent system with a working handoff and a human review gate that blocks the run.

TOOLCHAIN

Copilot Studio (connected agents), agent flows, Microsoft Teams approvals

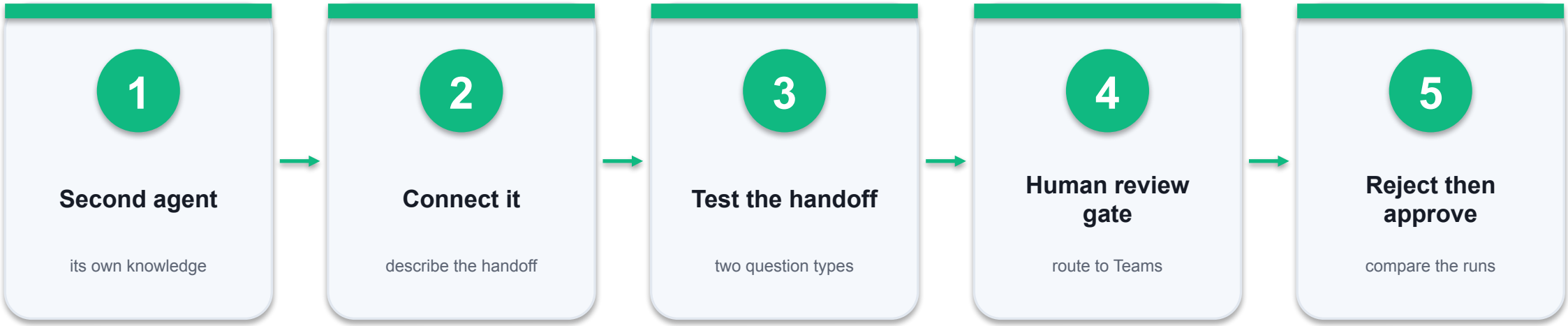
DONE WHEN

The claims question is answered by the connected agent, and a rejected booking leaves no Dataverse row while an approved one does.

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LAB 20 · PROCESS MAP

How Lab 20 Runs



THE POINT

A5/A7 — Compose connected agents and place a human where the consequence lands.

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LAB 20 · VERIFY

Multi-Agent Handoff and a Human Review Gate

✅ Expected result

The claims question is answered by the connected agent, and a rejected booking leaves no Dataverse row while an approved one does.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-20/README.md.

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LEARNING UNIT RECAP · YOU CAN NOW

Recap — Create and Manage Automated Processes with Power Automate and Copilot Studio Agents (1/2)

1

Lab 13 — Create Cloud Flows — Template, Copilot and Blank

A3/A5 — Build flows three ways and verify each run against its trigger and actions.

2

Lab 14 — Build an Approval Flow with a Human in the Loop

A3/A5 — Pause an automated process for a human decision and branch on the outcome.

3

Lab 15 — Button Flows and Scheduled Flows

A3 — Use manual and scheduled triggers, and collect user input at run time.

4

Lab 16 — Trigger Filters and Cross-Source Integration

A5/A7 — Stop unnecessary runs with trigger filters and highlight performance issues.

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LEARNING UNIT RECAP · YOU CAN NOW

Recap — Create and Manage Automated Processes with Power Automate and Copilot Studio Agents (2/2)

1

Lab 17 — Build Your First Copilot Studio Agent

A1/A3 — Build a conversational agent with instructions and knowledge as a new integration surface.

2

Lab 18 — Give the Agent a Tool — Calling a Workflow from an Agent

A4 — Expose a deterministic flow to an agent as a callable tool with typed inputs and outputs.

3

Lab 19 — Call an Agent from a Workflow with Structured Output

A4/A5 — Use an Agent node inside a flow and consume its typed fields downstream.

4

Lab 20 — Multi-Agent Handoff and a Human Review Gate

A5/A7 — Compose connected agents and place a human where the consequence lands.

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4

LU4

Model-Driven Apps with Power Apps and Dataverse

Topics 15–19 · Labs 21–26

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15

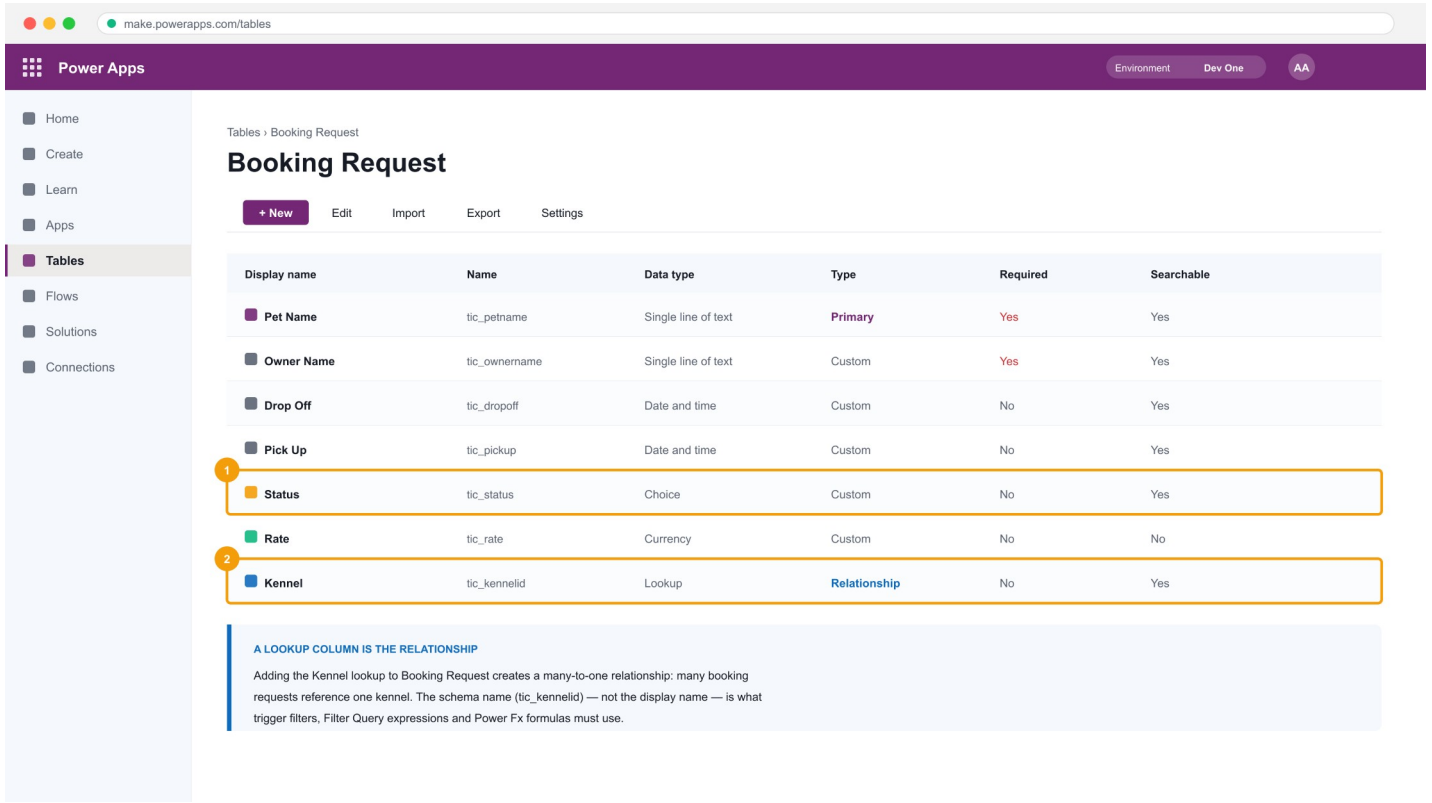
TOPIC 15

Create Tables in Dataverse

Table characteristics · properties · columns · relationships · keys

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A Dataverse Table Is More Than Rows and Columns



1. A Choice column stores a value from a defined option set — in a formula it is Status.Value.

2. A Lookup column IS the relationship. Adding it creates a many-to-one link to the other table.

Note the Name column: tic_kennelid. That schema name — not the display name — is what trigger filters, Filter Query expressions and Power Fx must use.

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The Parts of a Dataverse Table

1

Rows

Individual records — one booking, one property, one pet.

2

Columns

The metadata on each record, each with a data type.

3

Relationships

Connections between tables — one-to-many, many-to-one, many-to-many.

4

Keys

Unique identifiers; the primary key is a GUID the platform generates.

5

Primary column

The single text column that names the row across the product.

6

Views & forms

Saved queries and layouts that belong to the table itself.

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Choosing the Right Column Data Type

Data type	Stores	Watch out for
Single line of text	Short text up to 4,000 characters	Not for anything you will do maths on
Choice	One value from a defined option set	In formulas it is Status.Value, not Status
Lookup	A reference to a row in another table	This IS the relationship — it cannot be added later for free
Currency	Money, with a currency and exchange rate	Creates a hidden base-currency column too
Date and time / Date only	A moment, or a calendar date	Time-zone behaviour differs between the two
Whole number / Decimal	Counts and measurements	Decimal for money is a trap — use Currency

WHEN IT MATTERS Changing a column's data type after rows exist is often impossible — you create a new column and migrate. Model deliberately in Lab 21 before you build the app in Lab 22.

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16

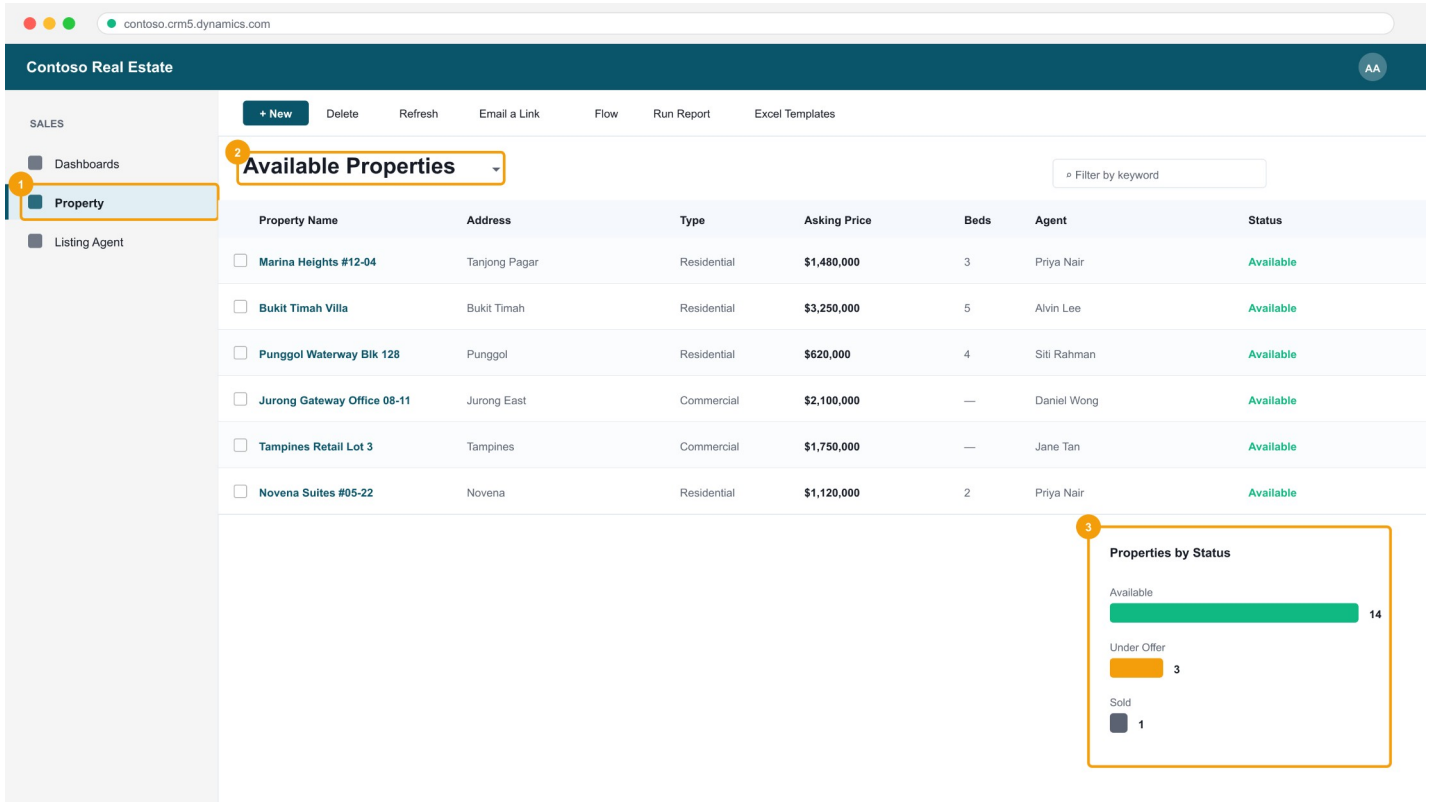
TOPIC 16

Get Started with Model-Driven Apps in Power Apps

Components · the app designer · site map · running the app

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A Model-Driven App Is Generated from the Data Model

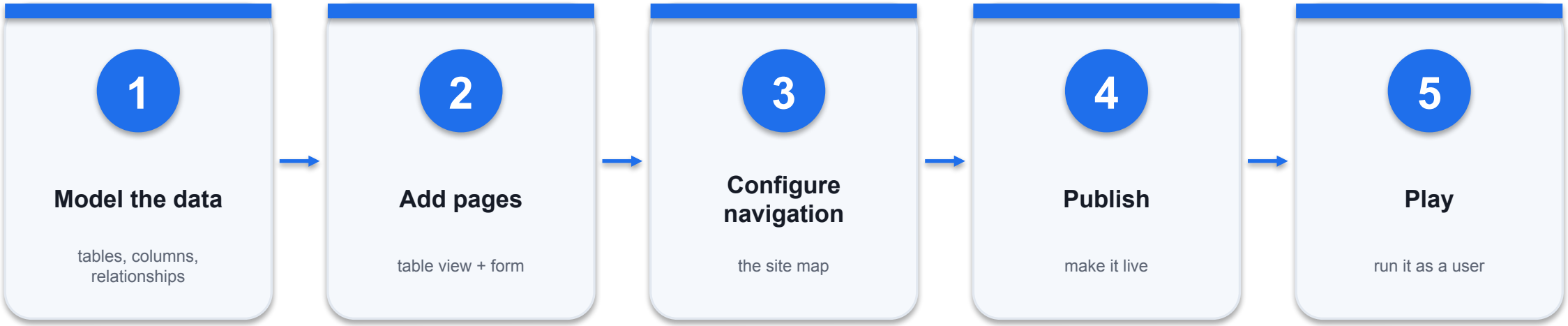


1. The site map — the navigation you configure from the tables in the app.
2. The view selector. Views are saved queries that belong to the table.
3. A chart, driven by the same Dataverse rows, with no code.

You did not draw this screen. You chose the tables, and the platform generated the grid, the forms, the commands and the search. Built in Labs 21–24.

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Building a Model-Driven App



THE DATA MODEL COMES FIRST

In a canvas app you can design the screen before you know the data. In a model-driven app you cannot — the app IS the data model, surfaced.

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17

TOPIC 17

Configure Forms, Charts and Dashboards

Form types · form elements · views · charts · dashboards

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The Four Form Types

Form type	Where it appears	Shows
Main	The record's own page	The full record, in tabs and sections
Quick View	Embedded in ANOTHER record's form	Read-only fields from a related row
Quick Create	A fly-out from + New	A short form for fast data entry
Card	In a view on a mobile layout	A few fields, compactly

WHEN IT MATTERS The Quick View form is how the Property record shows its agent's phone and email without the user navigating away — configured in Lab 23.

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Views, Charts and Dashboards

1

Public view

A saved query everyone can select — filter, sort and columns.

2

Personal view

A user's own saved query, visible only to them.

3

Chart

A visualisation over one table, respecting the current view's filter.

4

Dashboard

Charts plus lists on one page, cross-filtering each other.

5

Security trimmed

Every view, chart and dashboard shows only rows the user may read.

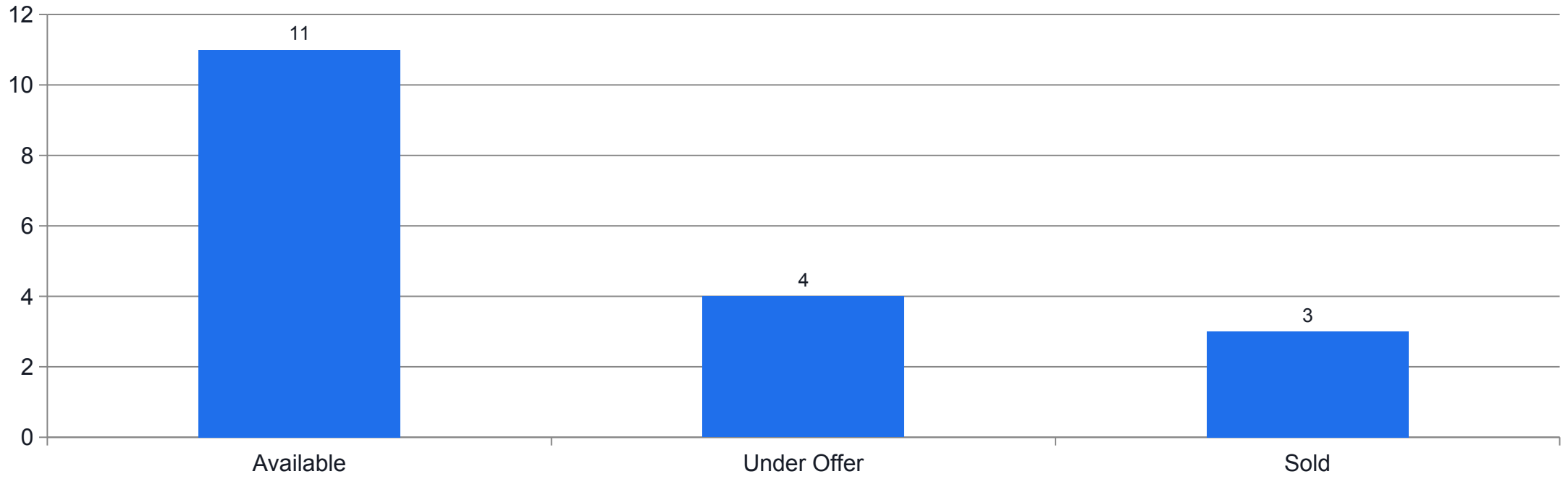
6

No code

All of it is configuration — this is the model-driven trade.

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What a Dataverse Chart Renders — Properties by Status

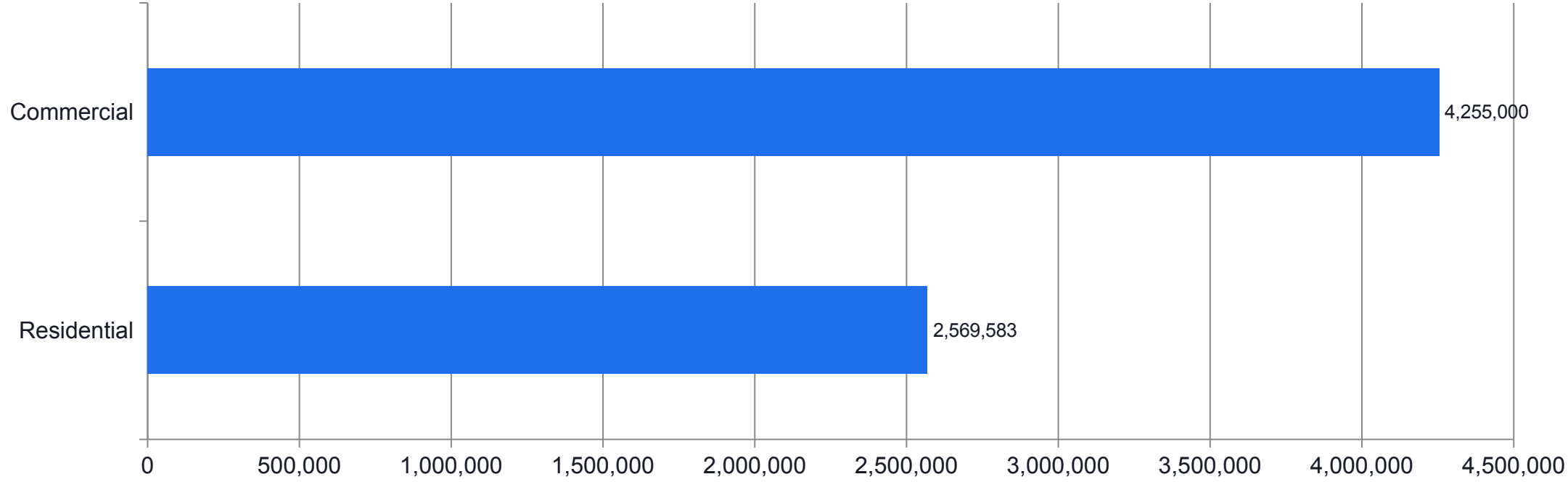


WHAT THE DATA SHOWS

These are the real counts from the 18 rows in Lab 21's properties.csv. The chart is a saved definition over the table, not a copy of the data — verify against expected-chart-values.csv in Lab 24.

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Average Asking Price by Property Type (SGD)



WHAT THE DATA SHOWS

Computed from the same 18 rows: 12 residential averaging SGD 2,569,583 and 6 commercial averaging SGD 4,255,000. An aggregate chart computes server-side across every row the USER may see — so two users can open the same chart and correctly see different numbers.

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18

TOPIC 18

Manage Solutions in Power Apps and Power Automate

Solutions · publishers · managed vs unmanaged · environments · ALM

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How a Change Reaches Production



THIS IS TSC A8

Implementing modifications to middleware safely means shipping them as a versioned, reviewable package — not editing production by hand.

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Managed or Unmanaged — The Decision That Cannot Be Undone

	Unmanaged	Managed
Purpose	Development	Distribution
Can components be edited?	Yes	No — locked in the target
Uninstall	Leaves components behind	Removes its components cleanly
Where it belongs	Your DEV environment	TEST and PRODUCTION
Reversible?	—	You cannot convert managed back to unmanaged

WHEN IT MATTERS Importing an unmanaged solution into production is the mistake that cannot be cleanly undone. Develop unmanaged, ship managed — Lab 25.

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What Belongs in the Solution

1. **Tables and columns**
   The data model your app depends on.
2. **Apps**
   Canvas and model-driven apps alike.
3. **Flows**
   Ship the automation WITH the app that relies on it.
4. **Forms, views, charts, dashboards**
   They belong to their tables.
5. **Connection references**
   So the target environment can rebind its own connections.
6. **Environment variables**
   Values that legitimately differ per environment.

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TOPIC 19

Guided Project & Troubleshooting Power Platform Integrations

Contoso Real Estate · isolating the failing layer · TSC A7 and A8

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Troubleshooting — Isolate the Layer Before You Change Anything



Loads for others → it is PERMISSION or ENVIRONMENT

Check the security role, the connection reference, and which environment the app or flow actually points at.

Fails for everyone → it is the FORMULA, the CONNECTOR or the DATA

Check delegation, the connector's status and licence, and whether the rows genuinely match the filter.

Lab 26 seeds four faults — a delegation warning, a failed flow action, a permission error and a broken connection reference — and you record layer, evidence and fix for each in troubleshooting-log.csv.

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Reading the Error — What Each Symptom Actually Means

Symptom	Layer	First thing to check
Blue delegation warning	Formula	A non-delegable function — rewrite it, don't raise the row limit
403 / 'permission'	Security	The user's Dataverse security role, not the app share
404 / 'not found'	Data	The record id you passed — read the action's Inputs
429 / throttled	Performance	Trigger filters and loop concurrency — you are running too often
Flow never triggers	Trigger	Column/row filters, the scope, and the environment
Agent answers wrongly	Grounding	General knowledge on, or 'Search all websites' still attached

WHEN IT MATTERS Every one of these appears in Lab 26. The discipline is the same each time: name the layer, capture the evidence, apply one change, re-test.

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WHAT YOU'LL DO

Hands-On Labs — Create and Manage Model-Driven Apps with Power Apps and Dataverse

Labs 21–22

- Create Dataverse Tables for a Model-Driven App
- Build a Model-Driven App

Labs 23–24

- Configure Forms and Views
- Create Charts and Dashboards

Labs 25–26

- Manage Solutions and Move to Another Environment
- Troubleshoot and Modify Power Platform Integrations

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LU4 · HANDS-ON

Create Dataverse Tables for a Model-Driven App

LAB 21

A1/A3 — Model the business domain as Dataverse tables, columns and relationships.

Working the Contoso Real Estate scenario, create Property and Listing Agent tables with typed columns, a choice column and a lookup relationship, then seed them with data.

YOU'LL BUILD

A two-table Dataverse data model with a relationship and seeded property records.

TOOLCHAIN

Power Apps, Microsoft Dataverse

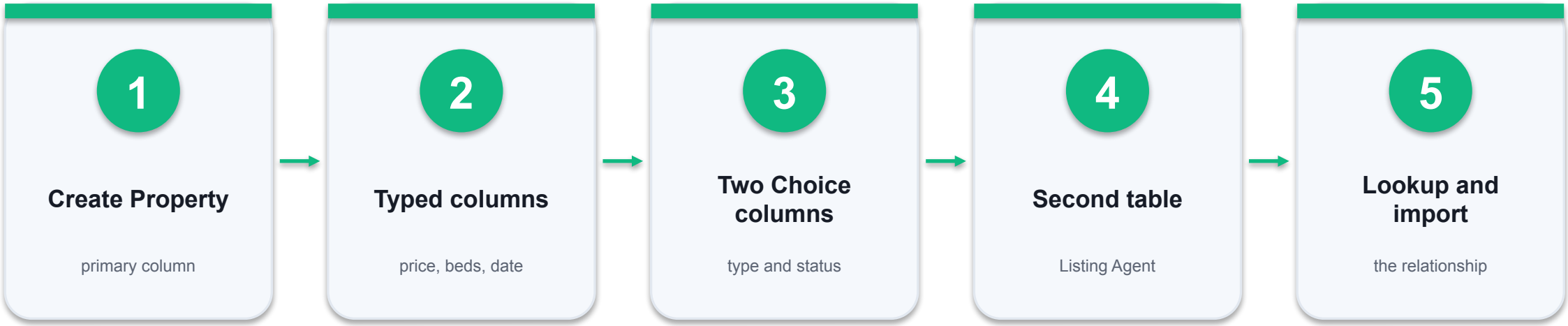
DONE WHEN

Both tables show imported rows, the choice columns render as dropdowns, and each Property row links to a Listing Agent.

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LAB 21 · PROCESS MAP

How Lab 21 Runs



THE POINT

A1/A3 — Model the business domain as Dataverse tables, columns and relationships.

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LAB 21 · VERIFY

Create Dataverse Tables for a Model-Driven App

✅ Expected result

Both tables show imported rows, the choice columns render as dropdowns, and each Property row links to a Listing Agent.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-21/README.md.

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LU4 · HANDS-ON

Build a Model-Driven App

LAB 22

A3 — Generate an application from the data model rather than drawing screens.

Create a model-driven app in the modern app designer, add the Property and Listing Agent tables to its navigation, configure the site map, and run the app.

YOU'LL BUILD

A running model-driven app with a two-area site map over the property data model.

TOOLCHAIN

Power Apps, Microsoft Dataverse, modern app designer

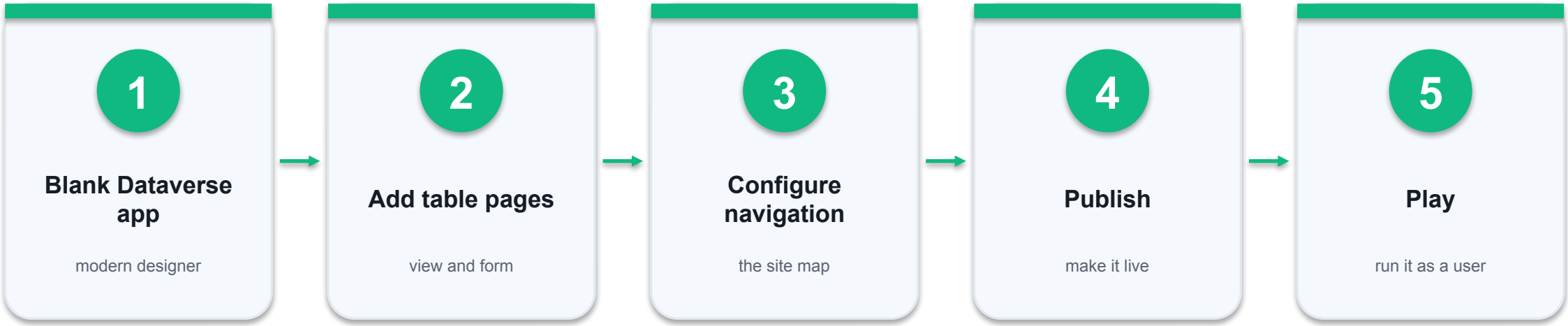
DONE WHEN

The published app opens with Property and Listing Agent in the navigation, and a record created in the app appears in the Dataverse table.

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LAB 22 · PROCESS MAP

How Lab 22 Runs



THE POINT

A3 — Generate an application from the data model rather than drawing screens.

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LAB 22 · VERIFY

Build a Model-Driven App

✅ Expected result

The published app opens with Property and Listing Agent in the navigation, and a record created in the app appears in the Dataverse table.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-22/README.md.

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LU4 · HANDS-ON

Configure Forms and Views

LAB 23

A3/A6 — Control what each user sees and edits, and verify the layout across the app.

Customise the Property main form with tabs, sections and a quick view of the related agent, then build filtered and sorted public views the app's users can select.

YOU'LL BUILD

A customised main form with a quick view control, plus two custom public views.

TOOLCHAIN

Power Apps, Dataverse, form and view designers

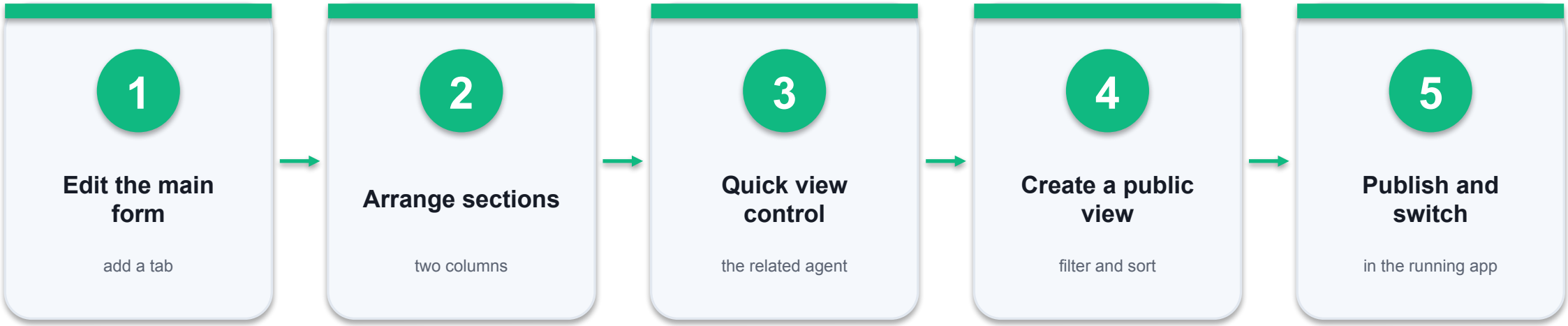
DONE WHEN

The main form shows both tabs and the agent's details in the quick view, and both custom views appear in the app's view selector with the correct rows.

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LAB 23 · PROCESS MAP

How Lab 23 Runs



THE POINT

A3/A6 — Control what each user sees and edits, and verify the layout across the app.

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LAB 23 · VERIFY

Configure Forms and Views

✅ Expected result

The main form shows both tabs and the agent's details in the quick view, and both custom views appear in the app's view selector with the correct rows.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-23/README.md.

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LU4 · HANDS-ON

Create Charts and Dashboards

LAB 24

A6 — Visualise integrated data to verify the platform is reporting correctly.

Build charts over the Property table and assemble them into an interactive dashboard inside the model-driven app.

YOU'LL BUILD

Two Dataverse charts and a dashboard surfaced in the model-driven app.

TOOLCHAIN

Power Apps, Dataverse charts and dashboards

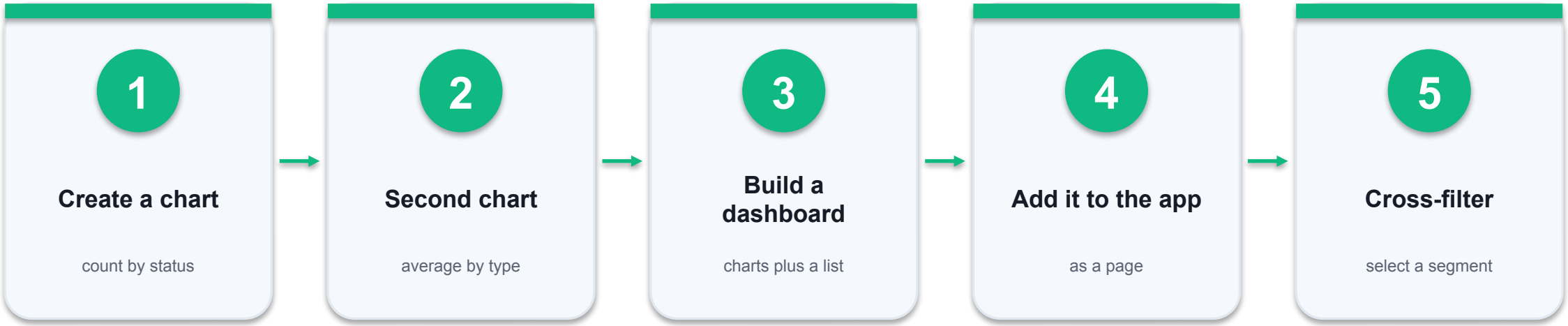
DONE WHEN

Both charts render real counts and averages, and selecting a chart segment in the dashboard filters the associated list.

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LAB 24 · PROCESS MAP

How Lab 24 Runs



THE POINT

A6 — Visualise integrated data to verify the platform is reporting correctly.

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LAB 24 · VERIFY

Create Charts and Dashboards

✅ Expected result

Both charts render real counts and averages, and selecting a chart segment in the dashboard filters the associated list.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-24/README.md.

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LU4 · HANDS-ON

Manage Solutions and Move to Another Environment

LAB 25

A8 — Package and ship middleware modifications between environments safely.

Create a publisher and an unmanaged solution, add the app, tables and flows to it, export it as managed, and import it into a second environment.

YOU'LL BUILD

A managed solution package exported from Dev and imported into a second environment.

TOOLCHAIN

Power Apps, Dataverse solutions, Power Platform admin centre

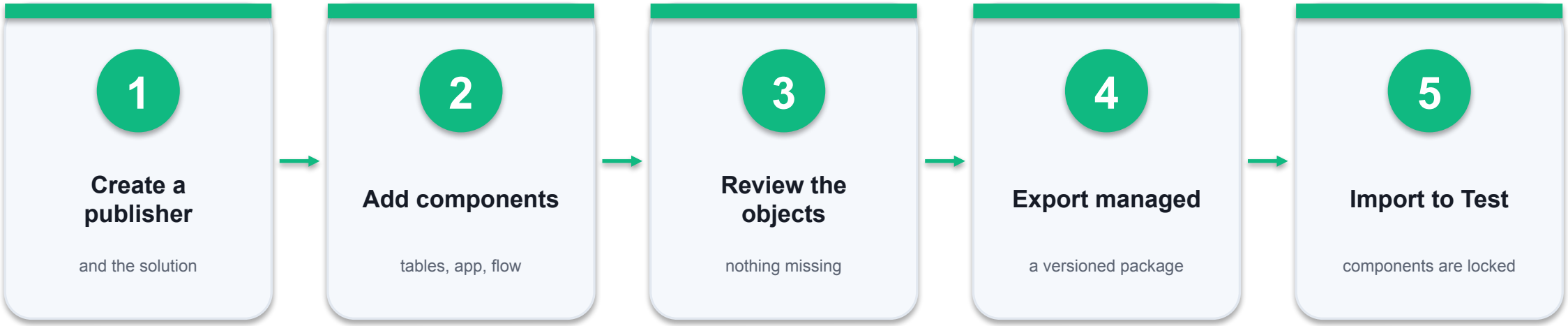
DONE WHEN

The managed solution imports into the second environment without errors and its components cannot be edited there.

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LAB 25 · PROCESS MAP

How Lab 25 Runs



THE POINT

A8 — Package and ship middleware modifications between environments safely.

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LAB 25 · VERIFY

Manage Solutions and Move to Another Environment

✅ Expected result

The managed solution imports into the second environment without errors and its components cannot be edited there.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-25/README.md.

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LU4 · HANDS-ON

Troubleshoot and Modify Power Platform Integrations

LAB 26

A7/A8 — Diagnose integration failures by layer and implement the corrective modification.

Work a set of seeded faults — a delegation warning, a failed flow action, a permission error and a broken connection reference — isolating each to its layer and applying the fix.

YOU'LL BUILD

A troubleshooting log recording each fault, the layer it belonged to, the evidence and the fix applied.

TOOLCHAIN

Power Apps Monitor, Power Automate run history, Power Platform admin centre, Dataverse

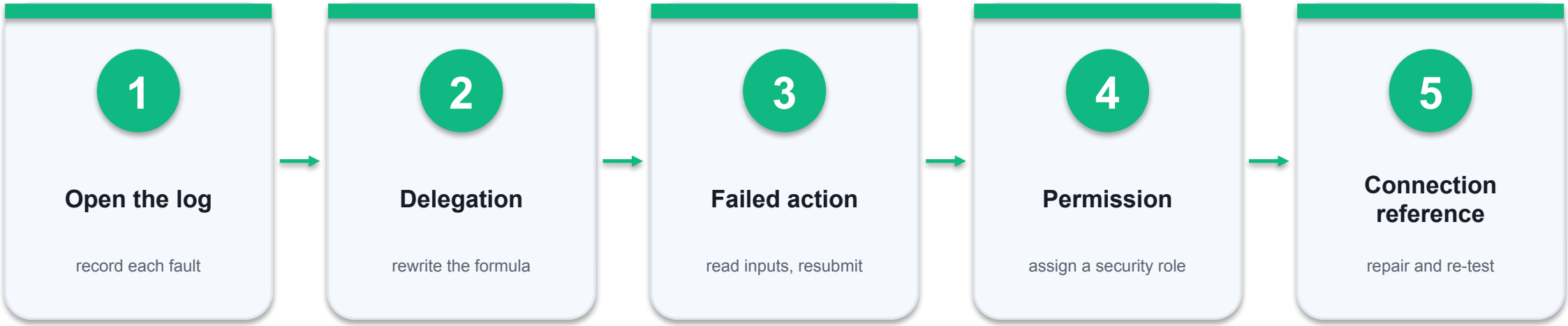
DONE WHEN

All four faults are resolved — no delegation warning, the flow run succeeds on resubmit, the shared user can open the app, and the connection reference is repaired — with each recorded in the log.

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LAB 26 · PROCESS MAP

How Lab 26 Runs



THE POINT

A7/A8 — Diagnose integration failures by layer and implement the corrective modification.

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LAB 26 · VERIFY

Troubleshoot and Modify Power Platform Integrations

✅ Expected result

All four faults are resolved — no delegation warning, the flow run succeeds on resubmit, the shared user can open the app, and the connection reference is repaired — with each recorded in the log.

IF IT DOESN'T WORK

Nothing appears

Confirm you are in the environment the trainer named — apps, flows and agents must share one.

Permission or 403 error

Sharing an app does not share its data. Check the Dataverse security role.

Step-by-step needed

Every numbered step, with the exact click path, is in the Learner Guide and labs/lab-26/README.md.

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LEARNING UNIT RECAP · YOU CAN NOW

Recap — Create and Manage Model-Driven Apps with Power Apps and Dataverse

1

Lab 21 — Create Dataverse Tables for a Model-Driven App

A1/A3 — Model the business domain as Dataverse tables, columns and relationships.

2

Lab 22 — Build a Model-Driven App

A3 — Generate an application from the data model rather than drawing screens.

3

Lab 23 — Configure Forms and Views

A3/A6 — Control what each user sees and edits, and verify the layout across the app.

4

Lab 24 — Create Charts and Dashboards

A6 — Visualise integrated data to verify the platform is reporting correctly.

5

Lab 25 — Manage Solutions and Move to Another Environment

A8 — Package and ship middleware modifications between environments safely.

6

Lab 26 — Troubleshoot and Modify Power Platform Integrations

A7/A8 — Diagnose integration failures by layer and implement the corrective modification.

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WRAP-UP

Course Summary & Assessment

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LEARNING OUTCOMES

What You Achieved

1

LO1 · Identified & scanned

Identified integration opportunities across Power Platform and recorded a connector feasibility scan (Lab 1).

2

LO2 · Integrated with Power Apps

Built canvas apps over Excel, Dataverse and SharePoint, wrote data with Patch and forms, and extended the platform with a PCF code component (Labs 5–12).

3

LO3 · Tested with Power Automate

Built flows on all four trigger families, added approvals and trigger filters, and verified every connection from run history (Labs 13–20).

4

LO4 · Resolved & modified

Diagnosed delegation, permission, connector and data faults by layer, and shipped modifications as a managed solution (Labs 25–26).

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WHAT TO TAKE BACK TO WORK

The Habits Worth Keeping

1

One environment

Apps, flows and agents must share it, or they cannot see each other.

2

Insert, never type

Dynamic content and the ⚡ picker — a typed value is tomorrow's bug.

3

Schema names, not display names

Trigger filters and Filter Query expressions need tic_status, not Status.

4

Put the control where the model cannot reach

A tool the agent lacks is a control; an instruction is a request.

5

Read the run history first

The first failure is the cause; everything after it is a symptom.

6

Ship managed, develop unmanaged

Importing unmanaged into production cannot be cleanly undone.

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NEXT STEPS

Continue Your Learning

1

Microsoft Learn

learn.microsoft.com/training — the PL-7001, PL-7002 and PL-7003 learning paths.

2

Power Apps samples

github.com/microsoft/PowerApps-Samples — PCF components, Dataverse and model-driven examples.

3

Rebuild from blank

Redo Labs 7, 14 and 18 without the guide until the pattern is automatic.

4

Extend the capstone

Add a Copilot Studio agent to the Contoso Real Estate app you built in LU4.

5

Certify

PL-900 Fundamentals, then PL-100 Power Platform App Maker.

6

Ship something real

Automate one process in your own organisation and measure the hours saved.

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CONTINUE WITH TERTIARY INFOTECH

Recommended Courses

1. **WSQ - Business Process Automation with Power Automate and Copilot Studio Agents (TGS-2022017524)**
2. **WSQ - Microsoft Power BI Data Analyst (PL-300)**
3. **WSQ - Microsoft Azure AI Fundamentals (AI-900)**
4. **WSQ - Develop Multi AI Agent Applications with Microsoft Copilot Studio**

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BEFORE YOU BEGIN

Briefing for Assessment

1

Phones and materials away

Place phones and other materials under the table or on the floor.

2

No photos or recording

Do not photograph or record any assessment script.

3

No discussion

No discussion of any kind during the assessment.

4

Black or blue pen

Use a black or blue pen for hard-copy scripts.

5

No correction fluid

No liquid paper or correction tape — strike through neatly instead.

6

Scripts collected on time

All scripts are collected when the time is up.

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Final Assessment

1

Written Assessment (WA)

Short-Answer Questions (SAQ) · 1 hour · open book.

2

Practical Performance (PP)

Hands-on Power Platform build tasks · 90 minutes · open book.

3

Open book means

Your slides, Learner Guide and approved course materials only.

4

Competency

You must be assessed Competent in BOTH the WA and the PP.

5

Attendance

A minimum of 75% attendance, per the SSG digital attendance record.

6

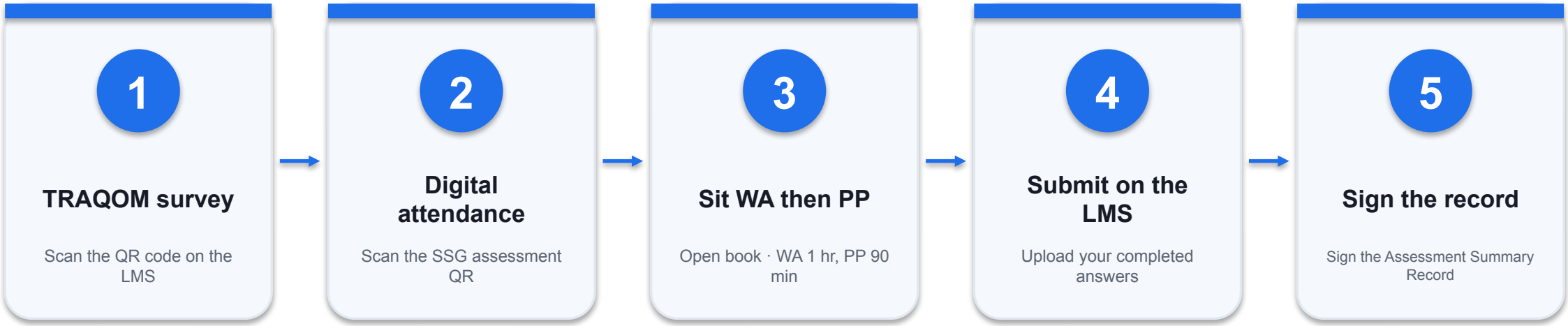
Appeal

If you disagree with the outcome, an appeal process is available — ask the trainer.

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ON ASSESSMENT DAY

Assessment Flow



REMEMBER

All five steps are mandatory for WSQ funding — missing the digital attendance or the TRAQOM survey can invalidate your claim.

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Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera (or Singpass app) and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

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WE'RE HERE TO HELP

Support

1

Email

enquiry@tertiaryinfotech.com

2

Telephone

+65 6100 0613

3

Website

www.tertiarycourses.com.sg

4

LMS / TMS

https://lms-tms.tertiaryinfotech.com

5

Post-course help

Free consultation on the course topics — just email us.

6

Your materials

Slides, Learner Guide, Lesson Plan and all 26 labs stay available on the LMS.

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HAPPY BUILDING

Thank You!

You can now build intelligent Power Apps, automate business processes with Power Automate, and ground and govern Copilot Studio agents — and defend where you put the human.

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